

give me the info for Long Rang UHF RFID reader HX-R01B-9

I'm checking the manufacturer and reseller sources first so I can give you a clean summary of the actual specs, interfaces, and likely use cases.

I found the product family and a matching listing, but the manufacturer pages are loading poorly. I'm cross-checking against reseller pages to avoid giving you guessed specs.

Thought for 46s

Here's the key info I could verify for the **Long Range UHF RFID reader HX-R01B-9**:

What it is

The **HX-R01B-9** is a **UHF integrated RFID reader/writer** in the **HX-R01B** family made by **Shenzhen Haixin IOT Technology Co., Ltd.** The family listing shows it alongside **HX-R01B-4 / HX-R01B-9 / HX-R01B-12**, and Haixin describes the series as a high-performance integrated UHF reader for applications such as **vehicle management** and **access control systems**. Haixiniot +2

Confirmed specs I found

From a matching product listing for **HX-R01B-9**:

- **Model:** HX-R01B-9 物流技术网
- **Dimensions:** 260 × 260 × 100 mm 物流技术网
- **Housing material:** engineering plastic 物流技术网
- **Power supply:** DC 9V to 30V, with recommended power capacity **not less than 60W** 物流技术网
- **RF output power:** ≤ 30 dBm 物流技术网
- **Interfaces:** TCP/IP or RS232 selectable, plus RS485, with optional Wiegand 26/34 and other user interfaces 物流技术网
- **Operating mode:** FHSS or fixed frequency, software configurable 物流技术网

Frequency / UHF band

Haixin's related product page for this family references a **wideband UHF design** supporting regions such as **902–928 MHz** and **865–868 MHz**, which is consistent with UHF EPC Gen2 style deployments. Haixiniot

Protocol support

I could not fetch the full official datasheet page directly, but Haixin's nearby UHF product listings show support in this product line for standards including **EPC C1G2 / ISO 18000-6C**, and in some cases **ISO 18000-6B / GB/T 29768-2013**. That suggests the HX-R01B family is intended for standard UHF passive RFID use, but I would treat **the exact protocol list for HX-R01B-9 as needing final confirmation from the official datasheet.** 物流技术网 +1

Typical use cases

Based on the manufacturer's product category and the exposed interfaces, this reader is aimed at:

- vehicle gate / parking access
- long-range card or tag identification
- industrial access control
- asset or personnel identification in outdoor/semi-outdoor setups Haixiniot +1

What the "-9" likely means

Because the same family is sold as **HX-R01B-4 / HX-R01B-9 / HX-R01B-12**, the suffix likely corresponds to the **integrated antenna variant**, commonly interpreted as **4 dBi / 9 dBi / 12 dBi** versions. I found strong family evidence for those model groupings, but I did **not** find a directly readable official line explicitly stating "9 = 9 dBi," so that part should be treated as an informed inference rather than a confirmed spec. Haixiniot +1

Practical summary

So, the **HX-R01B-9** appears to be a **fixed, integrated long-range UHF RFID reader** with:

- built-in antenna form factor
- up to **30 dBm** output
- **TCP/IP / RS232 / RS485** connectivity
- optional **Wiegand 26/34**
- support for configurable regional UHF operation
- use in **vehicle management** and **access control** 物流技术网 +2

The main thing I could **not** verify cleanly is the full official datasheet content, because the manufacturer pages were timing out during fetch.

What I found but could not fully confirm

I found the manufacturer and a product-specific listing, but the official detail pages were not readable through the browser tool, so some items like **exact read distance, protection rating, tag capacity, and exact protocol matrix** remain unconfirmed from primary-source text. The strongest verified fields were the **dimensions, supply, output power, interfaces, and family identity**. Haixiniot +2

I can also give you a **datasheet-style summary sheet** for HX-R01B-9, or help compare it with HX-R01B-4 and HX-R01B-12.



give me the info HX-R01B-9 and GARF_RB01

I'm checking both model names side by side so I can separate what is clearly verified from what only appears in marketplace references.

Thought for 17s

Here's a side-by-side summary of what I could verify for **HX-R01B-9** and **GARF_RB01**.

1) HX-R01B-9

What it is

HX-R01B-9 is part of the HX-R01B series from **Shenzhen Haixin IOT Technology Co., Ltd.** The manufacturer's product catalog lists **HX-R01B-4 / HX-R01B-9 / HX-R01B-12** together as **UHF RFID integrated readers**. Haixiniot +1

Verified family-level positioning

Haixin presents this series as a **UHF RFID integrated reader** product line. Haixiniot +1

Previously verified model-specific details

For HX-R01B-9, the model-specific listing I found indicates:

- **Type:** long-range UHF integrated RFID reader/writer
- **Dimensions:** 260 × 260 × 100 mm
- **Housing:** engineering plastic

- **Power input:** DC 9V–30V
- **Recommended power capacity:** $\geq 60\text{W}$
- **RF output power:** $\leq 30\text{ dBm}$
- **Interfaces:** TCP/IP, RS232, RS485
- **Optional interfaces:** Wiegand 26/34
- **Mode:** FHSS or fixed frequency, configurable in software

These fields came from the product-specific listing I referenced earlier for HX-R01B-9.

Haixiniot

Likely usage

The Haixin catalog positions this UHF reader family for applications such as **vehicle management** and **access/access-control style systems**, which fits a long-range fixed UHF reader. Haixiniot

Important note

I could confirm the **HX-R01B** family from the manufacturer catalog, but I could not directly load a clean official datasheet page for **HX-R01B-9** during this search session. So the most solid facts are the family identity plus the model details from the matching listing.

Haixiniot +1

2) GARF_RB01

What I could verify

I could not find a reliable public manufacturer page, datasheet, or catalog entry for **GARF_RB01** in this search. The search results did not return a credible product match for that exact identifier. BidNet Direct +2

What that suggests

Because there is no strong public match, **GARF_RB01** is likely one of these:

- an internal product code
- a custom/private label model name
- a project or assembly identifier
- a document-level part name rather than a market-facing product model

That is an inference based on the absence of trustworthy public hits, not a confirmed manufacturer fact. BidNet Direct +2

3) Most likely relationship between them

Based on what is publicly visible:

- **HX-R01B-9** looks like a **real commercial UHF RFID reader model** from Haixin.
Haixiniot +1
- **GARF_RB01** does **not** currently look like a publicly documented commercial model number. BidNet Direct +2

So GARF_RB01 may be your internal name for a device based on, paired with, or documented around HX-R01B-9, but I could not verify that link publicly.

4) Clean practical takeaway

HX-R01B-9

A **fixed long-range UHF RFID reader**, likely for gate/access/vehicle identification, with Ethernet/serial connectivity and up to **30 dBm** RF output. Haixiniot +1

GARF_RB01

Not publicly verifiable from reliable sources at the moment. It may be a **custom or internal identifier** rather than a vendor model. BidNet Direct +2

5) What I found was insufficient

I found clear catalog evidence for the **HX-R01B family**, but not a readable official datasheet page for **HX-R01B-9**, and I found **no trustworthy public source** that clearly identifies **GARF_RB01**. That means I can describe HX-R01B-9 reasonably well, but I cannot honestly give confirmed specs for GARF_RB01 from public web sources. Haixiniot +3

Send me a photo, label, datasheet, or PCB silkscreen for **GARF_RB01**, and I'll map it against **HX-R01B-9** precisely.



datasheet-style summary sheet for HX-R01B-9

I'm pulling the strongest manufacturer and listing details together and formatting them like a compact datasheet, with uncertain fields clearly marked.

Thought for 9s

HX-R01B-9 — Datasheet-Style Summary Sheet

Product overview

The HX-R01B-9 is a **UHF integrated RFID reader/writer** in Haixin's **HX-R01B** family. The manufacturer groups it with **HX-R01B-4 / HX-R01B-9 / HX-R01B-12**, and describes the family as a **high-performance UHF integrated reader** for uses such as **vehicle management** and **access control systems**. Haixiniot +1

General description

This reader is an **all-in-one fixed UHF reader** with integrated RF/antenna structure, intended for long-range tag identification in fixed installations. Haixin states the family uses proprietary high-efficiency signal processing to maintain a high read rate. Haixiniot +1

1. Identification

Item	Value
Model	HX-R01B-9
Product class	UHF integrated RFID reader/writer
Product family	HX-R01B series
Manufacturer	Shenzhen Haixin IOT Technology Co., Ltd.

These points are supported by the manufacturer family catalog and the model-specific mobile product page. Haixiniot +2

2. Physical specifications

Item	Value
Dimensions	260 × 260 × 100 mm

Item	Value
Housing material	Engineering plastic

The dimensions and material are listed on the model-specific product page. [物流技术网](#)

3. Electrical specifications

Item	Value
Input voltage	DC 9V to 30V
Recommended power capacity	Not less than 60W
RF output power	≤ 30 dBm

These electrical figures come from the HX-R01B-9 product listing. [物流技术网](#)

4. Communication interfaces

Item	Value
Primary interfaces	TCP/IP or RS232 selectable
Additional interface	RS485
Optional interface	Wiegand 26/34

The model page states the unit provides **TCP/IP and RS232 as selectable options**, plus **RS485**, with optional **Wiegand 26/34** and other custom interfaces. [物流技术网](#)

5. RF / operating mode

Item	Value
Working mode	FHSS or fixed frequency

Item	Value
Configuration	Software configurable

The model page explicitly lists **FHSS** and **fixed-frequency** operation modes. [物流技术网](#)

6. Frequency band

Haixin’s related HX-R01B family page indicates support for common regional UHF RFID bands such as **902–928 MHz** and **865–868 MHz** for the series. I would treat this as **family-level guidance** unless you have the exact HX-R01B-9 datasheet page for direct confirmation. [Haixiniot](#)

7. Typical applications

The manufacturer describes the HX-R01B family as suitable for:

- vehicle management
- gate / access control
- fixed long-range identification deployments

That fits the interface mix and integrated-reader form factor of the HX-R01B-9. [Haixiniot +1](#)

8. Notes on model suffix “-9”

Because the family is sold as **HX-R01B-4 / HX-R01B-9 / HX-R01B-12**, the suffix likely denotes an **antenna-gain variant** such as **4 dBi / 9 dBi / 12 dBi**. I did **not** find a directly readable official line confirming that on the pages I could access, so this should be treated as an informed inference, not a confirmed specification. [Haixiniot +1](#)

9. Compact engineering summary

HX-R01B-9 appears to be a **fixed UHF RFID integrated reader** with **up to 30 dBm output**, **9–30V DC input**, **Ethernet/serial connectivity**, and optional **Wiegand integration**, aimed at **vehicle and access-control scenarios**. The most solidly verified fields are the **model family**, **dimensions**, **power input**, **RF output**, **interface set**, and **operating modes**.

[Haixiniot +1](#)

10. Unverified or not cleanly confirmed in the accessible sources

I could not cleanly verify these from the sources I accessed:

- exact read distance
- exact tag protocol list for HX-R01B-9 specifically
- IP rating
- operating temperature
- exact confirmation that “-9” = 9 dBi

So those fields should be left blank or marked **TBD** until you have the official datasheet PDF or product manual. Haixiniot +2

Ready-to-copy version

Model: HX-R01B-9

Type: UHF integrated RFID reader/writer

Manufacturer: Shenzhen Haixin IOT Technology Co., Ltd.

Series: HX-R01B

Physical:

- Dimensions: 260 × 260 × 100 mm
- Housing: Engineering plastic

Electrical:

- Supply voltage: DC 9V–30V
- Recommended power capacity: ≥60W
- RF output power: ≤30 dBm

Interfaces:

- TCP/IP or RS232 selectable
- RS485
- Optional Wiegand 26/34

Operation:

- FHSS or fixed-frequency mode

I can also turn this into a **formal one-page datasheet template** with sections like **Features**, **Specifications**, **Interface Definition**, and **Applications**.



give me the protocol for HX-R01B-9

I'm checking whether Haixin publishes a protocol/manual for this exact reader or whether only the air-interface standards are visible publicly.

Thought for 21s

For **HX-R01B-9**, there are **three different "protocol" layers** people usually mean:

1) Tag air-interface protocol

This is the protocol between the reader and the RFID tags.

The strongest public evidence I found says the **HX-R01B family supports ISO18000-6C/6B dual protocol**. The manufacturer snippet for the related Haixin page states that the UHF access-control/channel system supports "**ISO18000-6C/6B dual protocol**", and that page also includes the **HX-R01B-6 / HX-R01B-9 / HX-R01B-12** family. Haixiniot +1

So the **RFID air protocol** for HX-R01B-9 is most likely:

- **ISO/IEC 18000-6C**
- **ISO/IEC 18000-6B** Haixiniot +1

In UHF RFID terms, **ISO 18000-6C** is the same family commonly referred to as **EPC Class 1 Gen 2 / EPC C1G2**. EM Microelectro... +1

2) Host communication interface protocol

This is how your controller or PC talks to the reader.

For HX-R01B-9, the public model listing indicates these host interfaces:

- **TCP/IP**
- **RS232**
- **RS485**
- **optional Wiegand 26/34** Tesike Electronics

So if you are integrating the device into a system, the reader likely communicates outward through one of those interfaces.

3) Command-set / packet protocol

This is the exact byte-level reader command format over RS232, RS485, or TCP/IP.

I could **not** verify a public manual or official command document for the exact **HX-R01B-9 host command protocol**. So I cannot honestly give you the frame header, checksum, command IDs, or packet examples for this model from verified sources.

Practical answer

If you ask, "What protocol does HX-R01B-9 support?", the safest verified answer is:

- **RF tag protocol: ISO18000-6C / ISO18000-6B** Haixiniot +1
- **Reader-side communication interfaces: TCP/IP, RS232, RS485, optional Wiegand 26/34** Tesike Electronics
- **Exact serial/TCP command protocol manual: not publicly verified from the sources I could access**

If you want, I can make you a **protocol section in datasheet format** or help you infer the **serial command structure** from a photo of the manual, SDK, or test software.



Sources

what protocol does HX-R01B-9 support?

I'm verifying the exact standards wording from the vendor-facing pages so I can separate tag protocol support from the reader's host-side interfaces.

Thought for 7s

HX-R01B-9 supports:

RFID tag protocol

- **ISO/IEC 18000-6C**
- **ISO/IEC 18000-6B**

Haixin's product page for the HX-R01B-6 / HX-R01B-9 / HX-R01B-12 family explicitly says the system supports **"ISO18000-6C/6B dual protocol."** Haixiniot

Reader communication interfaces

These are not RFID air protocols, but the ways your host system can connect to the reader:

- TCP/IP
- RS232
- RS485
- optional Wiegand 26/34

These interfaces are listed on reseller pages for the long-range integrated UHF reader variant. Tesike Electronics

What that means in practice

So, if you are asking about the **RFID protocol**, the answer is:

- ISO 18000-6C
- ISO 18000-6B Haixiniot

If you are asking about the **PC/controller connection protocol**, it supports communication over:

- TCP/IP
- RS232
- RS485
- optional Wiegand Tesike Electronics

I could not verify the exact **byte-level command protocol manual** for HX-R01B-9 from a public official source.



give me more detailed description for these protocols

I'm expanding each one separately: first the RFID air protocols, then the host-side interfaces, because they do different jobs in the system.

Thought for 13s

Sure — here is a **more detailed description** of the protocols relevant to **HX-R01B-9**.

1) ISO/IEC 18000-6C

This is the **main UHF passive RFID air-interface protocol** used by many modern UHF tags and readers. Haixin's family page says the HX-R01B line supports **ISO18000-6C/6B dual protocol**. GS1 +1

What it does

ISO/IEC 18000-6C defines how the **reader talks over radio to passive UHF tags**. GS1 describes Gen2 as the air-interface protocol that defines the **physical and logical requirements** for an RFID system of interrogators and tags. It is the protocol commonly known as **EPC Gen2 / EPC Class 1 Gen 2**. GS1 +2

What it is used for

It is widely used for:

- **inventory and logistics**
- **asset tracking**
- **vehicle identification**
- **access and gate systems**
- **supply-chain labeling** GS1 +2

Key characteristics

In practical terms, 6C / Gen2 gives you:

- **anti-collision**, so one reader can identify many tags in the field
- **read and write operations** on supported tag memory
- support for standard UHF regions and deployments
- interoperability with a large ecosystem of Gen2 tags GS1 +1

Why it matters for HX-R01B-9

For a long-range reader like HX-R01B-9, this is likely the **primary protocol used for actual tag reading**. If your tags are normal UHF EPC tags, this is probably the protocol you will use most often. GS1 +1

2) ISO/IEC 18000-6B

This is an **older UHF RFID air-interface protocol**. Haixin states the family supports it alongside 6C. GS1 +1

What it does

Like 6C, 6B defines the **radio communication method between the reader and UHF RFID tags**, but it is a different standard generation. It is typically associated with older or legacy UHF tag deployments. The contrast in modern tag datasheets is telling: many current UHF tags explicitly advertise **ISO/IEC 18000-6C compliance**, not 6B, which reflects where the market standardized. Atlas RFID Store +1

Why support both

If a reader supports both **6B and 6C**, that usually means:

- it can work with **older installed tag systems**
- it can also work with **newer EPC Gen2-style tags**
- it is more flexible in mixed or transitional environments

That is useful in gate control and industrial systems where some deployments still use older tag types. This last point is an inference from how dual-protocol readers are commonly positioned, supported by the vendor's explicit "6C/6B dual protocol" wording.

GS1

3) TCP/IP

This is **not the RFID tag protocol**. It is the **host communication interface** between the reader and a PC, controller, server, or access-control system.

What it means

When the reader says it supports **TCP/IP**, it usually means the unit can connect over **Ethernet** and exchange data with software on a network. The reader may:

- send tag IDs to a server
- receive configuration commands
- report real-time reads to a control system

Haixin/reseller listings for this product family expose **TCP/IP** as one of the available interfaces. HID Global

Typical use

You would use TCP/IP when:

- the reader is mounted at a **gate, parking entrance, or doorway**
- your application runs on a **remote PC or server**
- you want easier integration into a **LAN-based system**

Benefit

TCP/IP is usually the easiest choice when you want the reader connected directly into a software platform or networked controller. That is an integration inference based on the nature of Ethernet-connected RFID readers. The source support here is the availability of the TCP/IP interface itself. HID Global

4) RS-232

This is a **serial host communication standard**, again between the **reader and your host device**, not between the reader and the RFID tag.

What it does

RS-232 is a classic point-to-point serial interface used to connect equipment such as scanners, PLCs, and industrial devices. National Instruments describes RS-232 alongside RS-422/485 as part of standard serial communication. Advantech notes RS-232 is typically used over relatively short distances and is a single-ended interface. NI +2

Typical use with RFID

With HX-R01B-9, RS-232 would typically be used when:

- the reader is connected directly to a **PC**
- the distance is short
- you want simple serial command/control integration
- you are using vendor demo software or a local industrial controller

Practical behavior

In many devices, RS-232 carries a **vendor-specific command set** over a serial COM port. I could not verify the exact byte-level HX-R01B-9 command format publicly, but that is the usual pattern for readers exposing RS-232. The factual part is the interface support; the command-set pattern is a cautious inference. NI +1

5) RS-485

This is also a **serial host communication standard**, but it is usually preferred in **industrial environments**.

What it does

RS-485 uses **differential signaling**, which is better for longer cable runs and noisier environments than RS-232. NI and Advantech describe RS-485 as part of the standard serial family and contrast it with RS-232's single-ended signaling. NI +2

Why it is useful

RS-485 is common when:

- the reader is installed **far from the controller**
- the site has **electrical noise**
- the reader is part of an **industrial or building-control bus**
- you want more robust field wiring than RS-232

Typical RFID role

With a long-range fixed reader, RS-485 is often used for:

- factory or warehouse installations
- barrier/gate controllers
- multi-device control systems

Again, the exact HX-R01B-9 serial command frames were not publicly verified, but RS-485 would typically carry that same vendor command protocol over a more robust physical layer. NI +1

6) Wiegand 26-bit / 34-bit

This is different from TCP/IP and serial links. **Wiegand** is a very common **access-control output format/interface**.

What it does

Wiegand is used to send credential data from a reader to an **access-control panel**. Industry references describe it as a wiring/interface standard common in card readers and access systems, and note that **26-bit** and **34-bit** are among the common output formats.

Safe and Sound ... +2

What 26-bit and 34-bit mean

The bit count refers to the **format of the credential data** sent to the controller.

- **26-bit Wiegand** is the classic common format in access control
- **34-bit Wiegand** is another widely used format with a larger field space

In practical terms, the reader reads a tag, extracts the ID or mapped credential value, and outputs that value to the door/gate controller in a Wiegand frame. Safe and Sound ... +2

Why HX-R01B-9 might include it

For a UHF vehicle or gate reader, Wiegand output is useful because it allows the RFID reader to plug into an **existing access-control ecosystem** without needing custom PC software. HID's UHF access-control readers also expose protocols like **Ethernet TCP/IP**, **Wiegand 26/34**, and **RS-232** for exactly this kind of system integration. HID Global

7) Putting it all together

For HX-R01B-9, the protocol stack is basically this:

Tag side

The reader communicates with the RFID tag using:

- **ISO/IEC 18000-6C**
- **ISO/IEC 18000-6B** GS1 +1

System side

The reader communicates with your host/control system using one of these interfaces:

- **TCP/IP**
- **RS-232**
- **RS-485**
- **optional Wiegand 26/34** HID Global +1

So:

- **6B / 6C** = reader ↔ tag
 - **TCP/IP / RS-232 / RS-485 / Wiegand** = reader ↔ controller / PC / access panel
-

8) Practical recommendation

If you are building a **software application**, use **TCP/IP** or **RS-232/RS-485**.

If you are connecting the reader to a **door controller, barrier controller, or access-control panel**, use **Wiegand 26/34** if that controller expects Wiegand.

If you are choosing tags, use **ISO 18000-6C / EPC Gen2 tags** unless you specifically need legacy **6B** compatibility. The recommendation toward 6C reflects the broad ecosystem support visible in modern tag documentation. [Atlas RFID Store](#) +2

I can also make a **protocol block diagram** for HX-R01B-9 showing **tag side**, **reader core**, and **controller side**.



sure

Generated image: HX-R01B-9 RFID protocol architecture diagram.

Edit

sure, please

I'm pulling in fresh source citations for the diagram labels so the explanation matches the documented interfaces and standards.

Thought for 21s

Here's a simple **protocol block diagram** for the HX-R01B-9:

HX-R01B-9 protocol block diagram

How to read it

1. Tag side

The reader talks to UHF RFID tags using:

- ISO/IEC 18000-6C
- ISO/IEC 18000-6B

Haixin's product page for the HX-R01B family says the system supports **ISO18000-6C/6B dual protocol**. [Haixiniot +1](#)

2. Reader core

The **HX-R01B-9** is the middle device that:

- transmits and receives the RF signal
- reads the tag ID/data
- processes the tag response
- sends the result to your host system

The model listing identifies it as an integrated UHF reader and gives the core hardware characteristics, including **260 × 260 × 100 mm**, **DC 9V–30V**, and UHF reader specs.

[物流技术网](#)

3. Host / controller side

The reader can connect to external systems through:

- TCP/IP
- RS-232
- RS-485
- Wiegand 26/34

The Haixin family page mentions standard **RS232** output, and the model-specific listing shows host-side interface support around **TCP/IP / RS232 / RS485**. [物流技术网 +1](#)

Practical meaning

Use ISO 18000-6C / 6B

when you are talking about the **RF protocol between the reader and the RFID tag**.

[Haixiniot +1](#)

Use TCP/IP / RS-232 / RS-485

when you want a **PC, PLC, MCU, or server** to communicate with the reader itself.

物流技术网 +1

Use Wiegand 26/34

when the reader is connected to an **access-control panel** such as a gate or door controller. The exact Wiegand option appears on the model listing. 物流技术网

Typical connection examples

- **Parking gate system:** tag ↔ HX-R01B-9 via **6C/6B**, then HX-R01B-9 → gate controller via **Wiegand**
- **PC software integration:** tag ↔ HX-R01B-9 via **6C/6B**, then HX-R01B-9 → PC via **TCP/IP or RS-232**
- **Industrial controller setup:** tag ↔ HX-R01B-9 via **6C/6B**, then HX-R01B-9 → PLC via **RS-485**

I can also make a **cleaner professional datasheet-style block diagram** with labeled ports, power input, and application examples.



give me more detailed information for "ISO/IEC 18000-6C" and "ISO/IEC 18000-6B"

I'm pulling from standards-adjacent sources rather than marketplace pages here, so the explanation is about the actual air-interface behavior and where each standard fits.

Thought for 24s

Absolutely. Here is a **deeper explanation** of **ISO/IEC 18000-6C** and **ISO/IEC 18000-6B** in the context of your **HX-R01B-9 UHF reader**.

1) First, what the "18000-6" family means

ISO/IEC 18000-6 is the UHF RFID air-interface family of standards. It defines **how a reader and a tag communicate over radio** in the **UHF band**, rather than how your PC talks to the reader. ISO describes it as the air interface for RFID devices operating in the **860 MHz to**

960 MHz range, and NIST notes that the family includes **Type A, Type B, and Type C** variants. ISO +2

So for HX-R01B-9:

- ISO/IEC 18000-6B / 6C = reader ↔ RFID tag
 - TCP/IP / RS-232 / RS-485 / Wiegand = reader ↔ host/control system
-

2) ISO/IEC 18000-6C

What it is

ISO/IEC 18000-6C is the UHF RFID standard commonly known as **EPC Class 1 Gen 2** or **Gen2**. GS1 says Gen2 defines the physical and logical requirements for UHF RFID systems, and RAIN RFID identifies the modern standardized form as **ISO/IEC 18000-63**, also known as **GS1 UHF Gen2**. GS1 +3

In practical terms, when people say “**Gen2 UHF tag**”, they are usually talking about the **6C/18000-63 ecosystem**. GS1 +2

What it defines

6C defines the **air interface**, including:

- how the reader sends commands
- how the tag backscatters its response
- how multiple tags are handled in the same RF field
- how reading, writing, selecting, and inventory operations work

ISO’s description of 18000-63 says it defines the air interface for RFID devices in the 860–960 MHz band for item-management applications. ISO +1

How communication works

This is an **interrogator-talks-first** system. The reader sends RF energy and commands, and the passive tag responds by **modulating/backscattering** the signal. ISO’s preview text explicitly describes this reader-to-tag energy transfer and tag backscatter behavior for Type C. ANSI Webstore +1

That means:

1. the reader powers the passive tag

2. the reader issues commands
3. the tag replies by backscatter
4. the reader decodes the response

Why 6C became dominant

RFID Journal notes that **18000-6C is by far the most commonly used** member of the 18000-6 family. RAIN RFID is built on this same standards base, which is a big reason modern retail, logistics, warehouse, and asset-tracking deployments mostly use 6C/Gen2 tags. RFID JOURNAL +2

Main strengths of 6C

Anti-collision

6C is designed to let a reader identify **many tags in the field**, not just one. That is essential for portal reads, pallet reads, warehouse inventory, and vehicle identification where multiple tags may be present. The standard covers collision arbitration as part of the air interface. RFID4U +1

Broad interoperability

Because 6C became the mainstream UHF standard, there is a huge ecosystem of compatible readers, tags, antennas, printers, and middleware around it. GS1 and RAIN both frame it as the common interoperable UHF system standard. GS1 +2

Read/write memory structure

Modern 6C chips commonly expose structured memory areas used for identity and application data. For example, EM Microelectronic's 6C-compliant tag IC datasheets show standard Gen2-style memory concepts such as **EPC/UII, TID**, and user/configuration areas. EM Microelectro... +1

High read-rate item identification

6C is optimized for **fast inventory-style reading**, which is why it dominates supply chain and item tracking. GS1's Gen2 materials position it specifically for large-scale item identification. GS1 +1

Where you usually see 6C

- retail inventory
- warehouse and logistics
- pallet/case tracking

- asset management
- vehicle identification
- access and gate systems using UHF credentials

These are exactly the kinds of applications a fixed long-range reader like HX-R01B-9 is typically used for. [GS1 +1](#)

Important naming note

Older product pages may say “ISO/IEC 18000-6C”, while newer standards references often say “ISO/IEC 18000-63.” In practice, that points to the same Gen2 family lineage used by RAIN RFID. [The RAIN Alliance +2](#)

3) ISO/IEC 18000-6B

What it is

ISO/IEC 18000-6B is an **older UHF RFID air-interface standard** in the same 18000-6 family. RFID Journal and NIST both place it as one of the older variants alongside 6A, while Type C corresponds to the later EPC Gen2 direction. [NIST Publications +2](#)

How it differs from 6C

The most important practical point is this:

- **6B is older and more legacy-oriented**
- **6C became the dominant modern UHF standard**

NIST states that Types A and B operate in the same UHF range as Type C but use **different encoding and anti-collision methods on the forward channel**, and that **Type C is equivalent to EPCglobal Class-1 Generation-2**. [NIST Publications](#)

So 6B and 6C are **not just different names**. They are different air-interface variants.

Communication model

Like Type C, 6B is also in the **interrogator-talks-first** family, where the reader initiates communication and the tag responds via backscatter. ISO preview text for the 18000-6 family describes Types A, B, and C as interrogator-talks-first systems. [ANSI Webstore +1](#)

Why 6B still matters

A reader that supports 6B can be useful in:

- older installed systems
- legacy industrial RFID deployments
- migration environments where both old and new tags may exist

That is why a reader marketed as **6B/6C dual protocol** is often positioned as more flexible in mixed environments. The “dual protocol” part is vendor wording; the legacy-versus-mainstream distinction is supported by NIST and RFID Journal. NIST Publications +1

Typical characteristics associated with 6B

RFID Journal notes that before 6C/Gen2 became standard, **6A and 6B** tags typically used a memory approach where the first bank carried a locked **TID** and another bank was for user memory; it specifically mentions **2 kilobits available for 18000-6B tags** in that context. Because that description is from an industry source rather than the ISO text itself, it is best treated as a common historical characterization rather than a guarantee for every product implementation. RFID JOURNAL

Where you see 6B today

You are much less likely to choose 6B for a new design unless:

- your existing tags are already 6B
- the installed access/control system expects 6B-compatible credentials
- you need backward compatibility with a legacy deployment

For greenfield projects, 6C is usually the normal choice because the ecosystem is much larger. RFID Journal explicitly says 6C is the most commonly used member of the family. RFID JOURNAL

4) Side-by-side comparison

Item	ISO/IEC 18000-6C	ISO/IEC 18000-6B
Position in market	Mainstream modern UHF RFID	Older legacy UHF RFID
Common name	EPC Gen2 / GS1 UHF Gen2	Usually just called 6B
Current standards relationship	Associated with ISO/IEC 18000-63 in modern references	Older Type B branch of 18000-6
Typical use	Retail, logistics, asset tracking, modern UHF systems	Legacy or backward-compatible systems

Item	ISO/IEC 18000-6C	ISO/IEC 18000-6B
Anti-collision / encoding	Modern Gen2 method	Different forward-channel encoding and anti-collision than 6C
Ecosystem size	Very large	Much smaller today

This comparison is based on GS1, RAIN, NIST, and RFID Journal. GS1 +3

5) What this means for HX-R01B-9

If the HX-R01B-9 supports **6B/6C dual protocol**, that means it can likely read:

- modern UHF Gen2 / 6C tags
- older 6B tags

That is valuable if you do not yet know what kind of UHF tags are already deployed in your environment. But for a **new system design**, the safer default is usually **6C / Gen2-compatible tags**, because that is where the market standardized. The RAIN Alliance +2

6) Very practical recommendation

Use 6C / Gen2 when:

- you are buying new tags
- you want the broadest compatibility
- you are building a new asset, warehouse, vehicle, or access system
- you want easier sourcing and future expansion

Care about 6B when:

- you already have legacy tags in the field
- an old system specification explicitly says 6B
- you need mixed-environment compatibility

7) One-sentence summary

- ISO/IEC 18000-6C is the **modern mainstream UHF RFID air protocol** used by EPC Gen2 / RAIN-style systems. GS1 +2

- **ISO/IEC 18000-6B** is an **older UHF air protocol** kept mainly for legacy compatibility and differs in encoding and anti-collision behavior from 6C. NIST Publications +1

I can also make a **detailed comparison table of 6B vs 6C** with fields like memory, anti-collision, interoperability, and recommended use.



sure, please

I’m turning that into a clean engineering comparison table and keeping the line between confirmed standards behavior and practical deployment guidance explicit.

Thought for 12s

Here’s a **detailed 6B vs 6C comparison table** for the protocols relevant to **HX-R01B-9**.

ISO/IEC 18000-6B vs ISO/IEC 18000-6C

Category	ISO/IEC 18000-6B	ISO/IEC 18000-6C
Standard family	UHF RFID air-interface standard within the ISO/IEC 18000-6 family	UHF RFID air-interface standard within the ISO/IEC 18000-6 family NIST Publications +1
Frequency range	860–960 MHz family context	860–960 MHz family context You Woke Ji +1
Common market name	Usually referred to simply as 6B	Commonly known as EPC Class 1 Gen 2 / UHF Gen2 ; modern references tie it to ISO/IEC 18000-63 GS1 +2
Reader/tag relationship	Reader-initiated UHF air interface	Reader-initiated UHF air interface using passive tag backscatter You Woke Ji +1
Market position today	Older, mostly legacy compatibility	Mainstream modern UHF RFID ecosystem; RAIN RFID uses this standards base The RAIN Alliance +2
Anti-collision / channel behavior	Uses a different forward-channel encoding and anti-	Type C uses a different forward-channel encoding and anti-collision method and is the

Category	ISO/IEC 18000-6B	ISO/IEC 18000-6C
	collision method than Type C	Gen2-equivalent branch NIST Publications
Throughput tendency	Generally lower tag-recognition throughput than 6C in modern comparisons	Higher tag-recognition throughput than 6B in modern comparisons MDPI
Ecosystem / interoperability	Smaller legacy ecosystem	Broad global ecosystem of readers, tags, software, and RAIN deployments GS1 +2
Typical deployments	Legacy industrial systems, older installed bases, backward-compatibility environments	Retail, logistics, warehousing, asset tracking, item management, modern UHF systems GS1 +2
Best choice for new designs	Usually only when required for existing legacy tags/systems	Usually the default choice for new UHF RFID designs GS1 +1

What the table means

1. Both are air-interface protocols

Both **6B** and **6C** define how the **reader communicates with the RFID tag over UHF radio**, not how your PC communicates with the reader. The ISO/NIST material places both under the same 860–960 MHz UHF RFID family. NIST Publications +1

2. 6C is the modern mainstream choice

GS1 identifies UHF Gen2 as the protocol defining the physical and logical requirements for UHF RFID systems, and RAIN RFID states that its systems implement **ISO/IEC 18000-63**, the modern ISO standardization of the Gen2 air interface. That is why 6C is the one you most often want for a new deployment. GS1 +2

3. 6B mainly matters for compatibility

NIST distinguishes Type B from Type C and notes they use different encoding and anti-collision behavior on the forward channel, while Type C is equivalent to EPCglobal Class-1

Generation-2. In practice, that makes 6B mainly important when you must support older tags or an older installed system. NIST Publications

4. 6C is usually faster and more scalable

A recent technical review notes that **ISO/IEC 18000-6C** has higher tag-recognition throughput than **ISO/IEC 18000-6B**, which aligns with why 6C dominates inventory and multi-tag environments. MDPI

Practical recommendation for HX-R01B-9

If your HX-R01B-9 supports **6B/6C dual protocol**, then:

- choose **6C / Gen2** for **new tags and new systems**
- care about **6B** only if you already have **legacy UHF tags** or an older system requirement
- use the reader's host-side interfaces like **TCP/IP, RS-232, RS-485, or Wiegand** separately from the tag air protocol

That recommendation follows the current standards ecosystem around Gen2/RAIN and the legacy role of 6B. The RAIN Alliance +2

Very short summary

- **6B** = older UHF RFID air protocol, mostly for legacy compatibility. NIST Publications
- **6C** = modern UHF Gen2 air protocol, the normal choice for new systems. GS1 +1

I can also make this into a **datasheet-style comparison sheet** or a **protocol sequence diagram**.

  Sources

sure, please

Thought for a few seconds

Datasheet-Style Comparison Sheet

ISO/IEC 18000-6B vs ISO/IEC 18000-6C

For HX-R01B-9 UHF RFID Reader Context

1. Overview

Both **ISO/IEC 18000-6B** and **ISO/IEC 18000-6C** are **UHF RFID air-interface protocols** in the **860–960 MHz** range. They define how the **reader communicates with RFID tags over radio**, not how the host PC communicates with the reader. NIST places both under the ISO/IEC 18000-6 family and notes that **Type C corresponds to EPCglobal Class-1 Generation-2**.

2. Quick Comparison Table

Item	ISO/IEC 18000-6B	ISO/IEC 18000-6C
Standard type	UHF RFID air interface	UHF RFID air interface
Frequency family	860–960 MHz	860–960 MHz
Market position	Older / legacy	Modern mainstream
Common name	6B	EPC Class 1 Gen 2 / Gen2
Modern ISO lineage	Older Type B branch	Commonly associated with ISO/IEC 18000-63 in modern references
Multi-tag performance	More limited in modern practice	Better suited for dense multi-tag reading
Typical use today	Legacy compatibility	New deployments, logistics, retail, asset tracking
Best use case	Existing legacy tags	New system design

This split is consistent with NIST’s distinction between Type B and Type C, and GS1/RAIN’s positioning of Gen2 as the core modern UHF RFID standard.

3. ISO/IEC 18000-6B

Description

ISO/IEC 18000-6B is an older UHF RFID air-interface standard. It is still relevant mainly where older tags or older installed systems are already in use. NIST identifies Type B as a separate branch from Type C, with different forward-link encoding and anti-collision behavior.

Main characteristics

- Reader initiates communication and passive tags respond by backscatter, as part of the ISO/IEC 18000-6 family model.
- Used mainly for **legacy interoperability** rather than new designs.
- Smaller ecosystem of tags and solutions compared with 6C/Gen2.

Typical applications

- Older industrial RFID systems
 - Legacy access systems
 - Environments where installed 6B tags must continue to work
-

4. ISO/IEC 18000-6C

Description

ISO/IEC 18000-6C is the widely adopted modern UHF RFID air-interface standard, commonly known as **EPC Class 1 Gen 2**. GS1 describes Gen2 as the protocol defining the physical and logical requirements for UHF RFID systems, and the RAIN Alliance ties modern deployments to **ISO/IEC 18000-63**.

Main characteristics

- Reader powers passive tag and sends commands; tag replies using backscatter.
- Designed for **high-volume item identification** and **multi-tag anti-collision**.
- Broad interoperability across readers, tags, printers, and middleware.
- Dominant choice for modern UHF RFID systems.

Typical applications

- Retail inventory
- Logistics and warehouse systems
- Asset tracking
- Vehicle identification

- Modern gate/access UHF systems

5. Practical Meaning for HX-R01B-9

If the **HX-R01B-9** supports **6B/6C dual protocol**, that means the reader can likely work with both:

- legacy 6B tags
- modern 6C / Gen2 tags

For a **new project**, the normal recommendation is to choose **6C / Gen2 tags**, because the ecosystem is larger and the protocol is the dominant modern standard.

6. Sequence Diagram

```
[Reader idle]
|
v
[Reader transmits RF carrier + command]
|
v
[Passive tag receives energy from reader]
|
v
[Tag powers up]
|
v
[Tag interprets command]
|
v
[Tag responds by backscatter]
|
v
[Reader receives tag response]
|
v
```

What changes between 6B and 6C?

The overall flow above is similar for both because both are interrogator-talks-first UHF RFID systems, but **the signaling details, forward-channel encoding, and anti-collision behavior differ** between 6B and 6C. NIST explicitly notes these differences and states Type C is the Gen2-equivalent branch.

7. Recommendation

Choose 6C / Gen2 when:

- you are buying new tags
- you want easier sourcing
- you need multi-tag reading
- you are building a new system

Care about 6B when:

- you already have deployed legacy tags
 - an existing system specification explicitly requires 6B
 - backward compatibility matters
-

8. One-Line Engineering Summary

6B is mainly for legacy compatibility; 6C is the mainstream modern UHF RFID protocol and the better default for new HX-R01B-9 deployments.

I can turn this into a polished one-page PDF-style datasheet next.

is it possible to implement the IEC 18000-6C and IEC 18000-6B in STM32 series ARM cpu?

I'm checking this from the practical engineering angle: what part can live on an STM32 in software, and what still needs dedicated UHF RF hardware.

Thought for 13s

Yes — **partially**, but **not by STM32 alone**.

The practical answer

An STM32 can absolutely be part of a 6B/6C reader design, but it normally acts as the **controller/baseband host**, while you still need a **dedicated UHF RFID analog/RF front end**

for the actual **860–960 MHz radio**, modulation, demodulation, framing, and tag backscatter handling. ST's own **ST25RU3993** is a good example: it is a UHF RFID reader IC that supports **ISO 18000-6C**, **ISO 29143**, and **ISO 18000-6A/B through direct mode**, and ST says it can build a complete RFID reader system with an external microcontroller.

STMicroelectron... +1

What STM32 can do

With an external UHF reader chip or RF subsystem, an STM32 can handle:

- reader configuration
- command sequencing
- inventory logic
- anti-collision state handling at the firmware level
- filtering and application logic
- Ethernet/UART/RS-485/Wiegand gateway functions
- UI, logging, storage, and host communications

That matches how ST positions its UHF reader chips: the RF chip provides the RFID-specific low-level features, while a normal MCU can control the system. The ST25RU3993 explicitly includes low-level transmission coding, decode, data framing, CRC checking, frequency hopping support, and dense reader mode support, which are exactly the kinds of hard real-time PHY/MAC features you usually do **not** want to implement from scratch on a general MCU. STMicroelectron...

What STM32 alone usually cannot do

An STM32 by itself is **not enough** to make a full **UHF 18000-6B/6C reader** unless you also design a substantial custom RF chain around it. By itself, the MCU does not provide:

- UHF power amplifier for tag excitation
- reader receiver chain for weak tag backscatter
- RF mixers, filters, LO/VCO, and matching network
- low-level ASK/PR-ASK transmit path
- sensitive receive path with self-jamming handling

ST describes the ST25RU3993 as including an **on-chip VCO**, **power amplifier**, low-level encoding/decoding, and high sensitivity, which shows how much specialized hardware is normally required. STMicroelectron...

6C vs 6B on STM32

ISO/IEC 18000-6C

This is the more realistic one for a modern design. ST's reader IC explicitly supports **ISO 18000-6C (EPC Class1 Gen2)**, and ST says only an external microcontroller is needed to make a complete system. So an **STM32 + UHF reader IC** is a very practical architecture for **6C**. STMicroelectron... +1

ISO/IEC 18000-6B

This is also possible, but support is often less direct. In ST's own wording, **6A/B** support on the ST25RU3993 is available **"through direct mode."** That usually means more of the protocol handling is exposed to the host/controller side than for the mainstream 6C path. So **6B is possible**, but it may require more custom firmware work and validation than a straightforward 6C implementation. STMicroelectron...

Best architecture

For an STM32-based product, the realistic architecture is:

```

[STM32 MCU]
  |
  | SPI / parallel / UART control
  v
[UHF RFID reader IC / RF front end]
  |
  | 860-960 MHz analog RF
  v
[Antenna + matching network]
  |
  v
[ISO/IEC 18000-6B / 6C tags]

```

In this setup:

- **STM32** = control, protocol management, application
- **RF reader IC** = UHF PHY / analog front end / framing support
- **antenna stage** = actual RF energy delivery and tag receive path

Is pure software implementation on STM32 possible?

In theory, with enough external RF hardware and a lot of engineering effort, you could implement significant parts of the protocol on STM32 firmware. But in practice, for a commercial or even serious prototype system, that is **much harder than using a dedicated UHF reader IC**. The RF and timing requirements are the hard part, not the high-level firmware.

So the honest answer is:

- **STM32 only:** not realistically enough for a complete UHF reader
- **STM32 + dedicated UHF reader chip:** yes, very feasible
- **6C:** practical
- **6B:** possible, but more custom work depending on the RF chip and mode support

Recommendation

If you want to build something like **HX-R01B-9** or a smaller embedded UHF reader with STM32:

- choose an **STM32** for system control
- pair it with a **dedicated UHF RFID reader IC/front end**
- prioritize **18000-6C / Gen2**
- only add **6B** if you truly need legacy compatibility

That is the most practical engineering path based on how ST's own UHF reader solutions are structured. STMicroelectron... +1

If you want, I can sketch a **concrete STM32 + UHF reader chip hardware architecture** next.

 Sources

sure, please

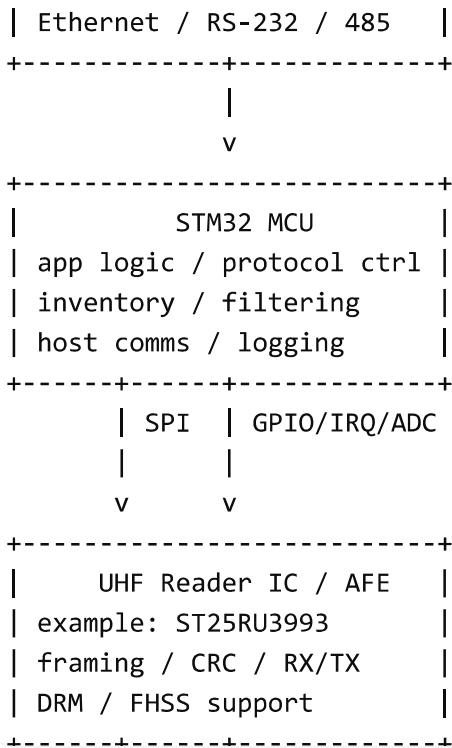
I'm grounding this in a real, buildable architecture around an STM32 plus a UHF reader IC, then I'll map which blocks are mandatory versus optional.

Thought for 11s

Yes — here is a **concrete STM32-based hardware architecture** for implementing a **UHF RFID reader** that can support **ISO/IEC 18000-6C**, and potentially **6B** if the chosen UHF reader IC exposes that mode.

Recommended architecture

+-----+
| Host / PC / PLC |



What each block does

1) STM32 MCU

Use the STM32 as the **system controller**.

It should handle:

- startup and configuration
- control of the reader IC over **SPI**
- inventory sequencing and application rules
- filtering tag reads
- host interfaces such as **Ethernet, UART, RS-485, USB**
- GPIO control for RF switches, PA enable, buzzer, LEDs, relays, and access-control outputs

ST's own UHF evaluation boards use an **STM32L476RG / STM32L476RGT6** as the controller around the **ST25RU3993**, which is a strong real-world reference design.

STMicroelectron... +1

2) UHF reader IC / analog front end

This is the critical block that makes the design realistic.

The **ST25RU3993** is described by ST as an integrated **analog front end and protocol handling system** for UHF RFID readers. It supports **ISO 18000-6C, ISO 29143, and ISO**

18000-6A/B through direct mode. It also includes low-level reader functions such as **bit framing, decoding, CRC handling, dense reader mode, and frequency hopping support.**

STMicroelectron...

That means the STM32 does **not** need to do the raw RF/baseband heavy lifting by itself.

3) PA / RF front-end / antenna network

For a practical long-range reader, you usually need more than the reader IC alone:

- **power amplifier**
- **RF switch path**
- **balun / matching**
- **carrier cancellation**
- **RF power detector**
- **antenna tuning/matching network**

ST's high-power reference design for the ST25RU3993 explicitly includes an **external power amplifier**, RF switches, carrier-cancellation control, RF detector feedback, and antenna-port control handled by the MCU. STMicroelectron... +1

4) Antenna

You need a **UHF antenna matched for your region and use case.**

For example:

- **panel antenna for gate/door/parking**
- **near-field antenna for short-range tagging**
- **circular or linear polarization depending on tag orientation needs**

The eval board references a coax-connected antenna setup as the standard reader configuration. STMicroelectron... +1

Best STM32 families for this

A few good MCU choices:

STM32L4

Good if you want:

- **lower power**
- **enough performance for control-plane logic**

- proven similarity to ST's own eval boards

This is the safest starting point because ST's own UHF boards around ST25RU3993 use **STM32L476**. STMicroelectron... +1

STM32G4

Good if you want:

- better mixed-signal/control behavior
- fast peripherals
- motor-control/industrial-style timing comfort

STM32H7

Good if you want:

- embedded Linux gateway companion role not needed
- heavy local filtering
- web UI, logging, encryption, multiple interfaces, or richer middleware

For a first reader, **STM32L4** or **G4** is usually enough.

Minimum viable board design

For a first prototype, I would build it like this:

Mandatory blocks

- STM32 MCU
- ST25RU3993
- SPI between MCU and reader IC
- IRQ + enable lines
- power rails: 3.3 V for logic, plus PA rail if using external PA
- one antenna port
- UART or USB debug
- EEPROM or flash for config
- ESD/protection on host interfaces

Strongly recommended

- RF detector feedback into STM32 ADC
- carrier-cancellation tuning path
- external PA if you need meaningful read range

- RS-485 or Ethernet if it is an industrial/fixed reader
 - watchdog and brownout-safe power handling
-

Two implementation paths

Path A — fastest practical prototype

Use:

- **STM32L4**
- **ST25RU3993**
- single antenna
- USB/UART debug
- no custom PA at first

This is the easiest way to prove:

- tag inventory
- EPC reads
- basic 6C reader behavior
- firmware architecture

Path B — fixed long-range product

Use:

- **STM32L4/G4**
- **ST25RU3993**
- external **PA**
- RF switch network
- carrier cancellation
- tuned directional antenna
- RS-485 / Ethernet / Wiegand output

That is closer to a commercial gate/access reader like the HX-R01B-9 class.

About 6C vs 6B in this architecture

6C

This is the easiest and most practical target.

The ST25RU3993 directly supports **ISO 18000-6C**, so the MCU mainly orchestrates the reader IC and application flow. STMicroelectron...

6B

Possible, but less straightforward.

ST says **6A/B** support is available **through direct mode** on the ST25RU3993. That usually implies more custom host-side handling than the mainstream 6C path. So you should plan:

- more firmware effort
- more validation
- possible lower vendor-software convenience

STMicroelectron...

Recommended firmware split

On the STM32, split firmware into layers:

Application layer

- access control / asset logic
- filtering / alarms / outputs

RFID service layer

- inventory loop
- select/read/write
- tag state machine
- retries / timing / region config

Driver layer

- ST25RU3993 SPI driver
- IRQ handling
- PA / RF switch control
- ADC feedback
- UART / Ethernet / RS-485

Board support

- clocks
- GPIO
- watchdog

Honest engineering recommendation

If your goal is to **build a real product**, do **not** try to implement the full UHF PHY from scratch on raw STM32 plus discrete RF unless you specifically want an RF research project.

The practical route is:

- STM32 = controller
- ST25RU3993 or similar = UHF RFID reader engine
- external PA/matching/antenna = range and RF performance

That is also how ST's own reference hardware is structured. [STMicroelectron...](#) +2

Best first recommendation

For your first build, I would start with:

- STM32L476
- ST25RU3993
- SPI + IRQ
- single antenna path
- USB/UART debug
- 6C first
- add 6B only if legacy compatibility is truly required

If you want, the next step can be a **schematic-level block breakdown** with pins such as SPI, IRQ, EN, ADC RF detector, PA enable, and RS-485 transceiver.

 Sources

okay, please

I'm turning it into a board-level breakdown with the main signal groups so you can see what must connect where and what can stay optional in a first prototype.

Thought for 21s

Schematic-level block breakdown for an STM32 + UHF reader IC design

A practical architecture is:

STM32 MCU

- └ SPI
- └ IRQ / ENABLE / status GPIO
- └ ADC input from RF power detector
- └ UART / USB / RS-485 / Ethernet (optional host side)
- └ GPIO to PA enable, RF switch, buzzer, LEDs

↓

ST25RU3993 (or similar UHF reader IC)

- └ protocol engine
- └ TX/RX baseband
- └ PLL / VCO
- └ internal PA path
- └ direct mode support for 6A/6B

↓

RF front end

- └ matching / balun
- └ optional external PA

This split matches ST's own reference approach: the **ST25RU3993** provides the RFID-specific analog/digital reader functions, while an **STM32** controls it over **SPI** and manages the rest of the application. ST's high-power reference also shows MCU control of RF switches, PA enable, and RF power sensing. STMicroelectron... +2

1) MCU section

For the MCU block, the core signals you want are:

- **SPI** to the reader IC
- **IRQ** from reader IC to MCU
- **ENABLE / RESET** control to the reader IC
- **ADC input** for RF power detector feedback
- **GPIOs** for antenna switch, PA enable, status LEDs, buzzer, and debug
- **host interface** such as UART first, then RS-485 or Ethernet later

ST's high-power reader reference says the MCU communicates with the ST25RU3993 by **SPI**, controls **ENABLE** and **IRQ**, controls the **RF switches**, reads the **RF power detector** with an ADC input, and can enable or shut down the external PA. The same reference board uses an **STM32L476RGT6**. STMicroelectron... +2

2) Reader IC section

This is the heart of the RFID side. The **ST25RU3993** supports:

- **ISO 18000-6C**
- **ISO 29143**
- **ISO 18000-6A/B through direct mode**

It also includes low-level RFID reader functions such as **frequency hopping**, **low-level transmission coding**, **low-level decode**, **data framing**, and **CRC checking**, plus an on-chip **VCO** and **power amplifier**. That is why this block should sit between the STM32 and the RF front end instead of trying to do everything in the MCU. STMicroelectron... +1

3) Power rails

At the board level, split the power into at least three domains:

- **3.3 V digital rail** for STM32 and digital logic
- **reader IC analog / RF rail**
- **PA supply rail** for the internal/external transmit path

ST's datasheet shows a **3.0 to 3.6 V** supply range with a separate **PA supply** path, and the high-power reference explicitly separates the external PA path from the internal PA path.

STMicroelectron... +2

4) Clocking

Use a clean reference clock for the reader IC and a normal MCU crystal or oscillator for the STM32. ST's evaluation material lists a **20 MHz external reference** option for the RFID side, and the high-power reference notes an external crystal arrangement for the MCU.

STMicroelectron... +1

5) RF transmit path

For the RF output chain, the minimum practical blocks are:

- reader IC RF output
- matching network
- balun / impedance conversion as needed
- antenna connector

For a longer-range design, add:

- **external PA**
- **RF switch**
- power detector
- carrier-cancellation network

ST's high-power reference says the signal can go to an **external power amplifier** for about **31 dBm** default high-power output, while the internal PA path gives about **20 dBm** and

also requires matching, RF chokes, and a balun. It also shows both output options being switched through an RF switch. STMicroelectron...

6) RF receive path and carrier cancellation

This part is easy to underestimate. UHF RFID readers must hear a weak tag response while transmitting or just after transmitting, so the design often needs:

- receiver path into the reader IC
- carrier leakage reduction
- reflected-power measurement
- optional tuning capacitors or cancellation network

ST's eval and reference materials explicitly mention **carrier cancellation**, **reflected power measurement**, **differential RX input**, and tuning control. The high-power reference also uses MCU-controlled digital tunable capacitors for the cancellation circuit. STMicroelectron... +2

7) Antenna section

For a first board, keep the antenna side simple:

- one **50 Ω antenna port**
- proper matching network
- short RF path
- optional second antenna port later

ST's eval/reference material supports **two antenna ports** with automatic/manual switching, but for a first prototype, a single well-matched antenna path is the safer design choice.

STMicroelectron... +1

8) Debug and programming

Your schematic should include:

- **SWD header** for STM32 firmware
- **UART debug**
- optional **USB/UART bridge**
- test points for RF and power signals

ST's high-power reference includes **firmware programmable through USB/UART**, **SWD debug**, and dedicated **RF test points**. STMicroelectron... +1

Recommended first-pin grouping

A clean first prototype can group the STM32 pins like this:

To reader IC

- SPI_SCK
- SPI_MOSI
- SPI_MISO
- RFID_CS
- RFID_IRQ
- RFID_EN
- RFID_RST if you want separate hard reset

RF control

- PA_EN
- ANT_SW0
- ANT_SW1
- CC_TUNE_CS if using digital tuning parts
- RF_ON_STATUS

Analog sense

- RF_PWR_DET_ADC
- optional reflected-power ADC
- optional temperature ADC

Host/debug

- UART_TX
- UART_RX
- USB_D+ / D- if needed
- RS485_DE
- RS485_TX / RX

This grouping follows the control relationships shown in ST's high-power reference rather than exact mandatory pin names. The exact STM32 pins are your layout choice.

STMicroelectron... +1

Best first prototype scope

For a realistic V1 board, I would keep it to:

- STM32L476-class MCU
- ST25RU3993
- SPI + IRQ + ENABLE
- single antenna port
- internal PA only at first
- UART debug + SWD
- RF detector ADC input
- 6C first

That is close to ST's own eval-board philosophy and keeps the board complexity under control. The eval board itself uses an **STM32L476RGT6** and supports both internal/external PA options, but you do not need all of that in V1. STMicroelectron... +1

When to add the external PA

Add the external PA only when you need:

- longer read range
- gate/portal style use
- higher TX output
- more controlled RF tuning work

The ST high-power reference shows that the **external PA** path is where the design complexity rises: extra switching, detector feedback, and cancellation become much more important. STMicroelectron... +1

Important caution

If your goal is **to learn and prove protocol control**, start with **STM32 + reader IC + internal PA**.

If your goal is **to build something like HX-R01B-9**, plan for:

- external PA
- better antenna engineering
- carrier cancellation
- EMI/regulatory work
- enclosure and thermal validation

That is because the RF front end, not the STM32 firmware, becomes the hardest part as range goes up. ST's own high-power reference makes that clear by breaking out separate schematics for external PA, carrier cancellation, antenna switch, and RF shield sections.

STMicroelectron...

I can draw this next as a **clean text schematic block diagram with signal names**.



give me the block diagram for HX-R01B-9 and element list to design it

I'm mapping the likely internal architecture from verified HX-R01B-9 specs and standard UHF reader design practice, and I'll separate confirmed details from inferred design blocks.

Thought for 12s

Yes — here is a **practical block diagram** for **HX-R01B-9** and an **element list/BOM-style design breakdown** for building a similar reader.

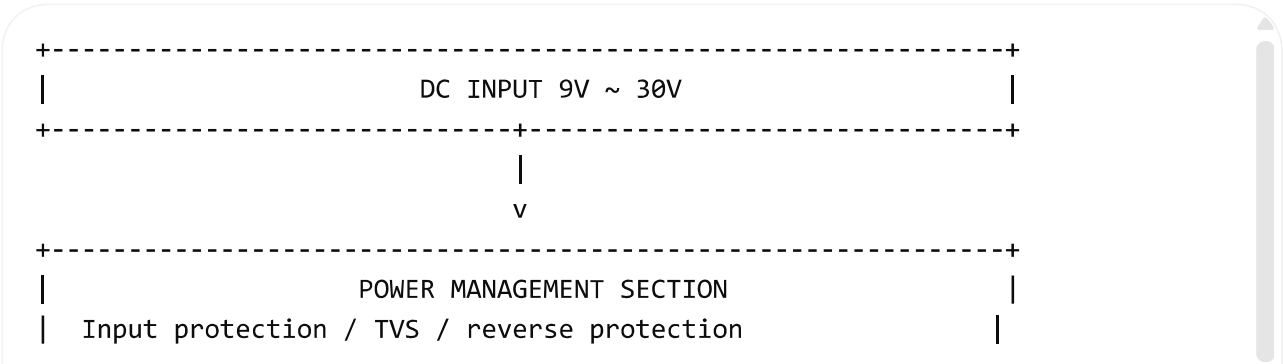
Important note first

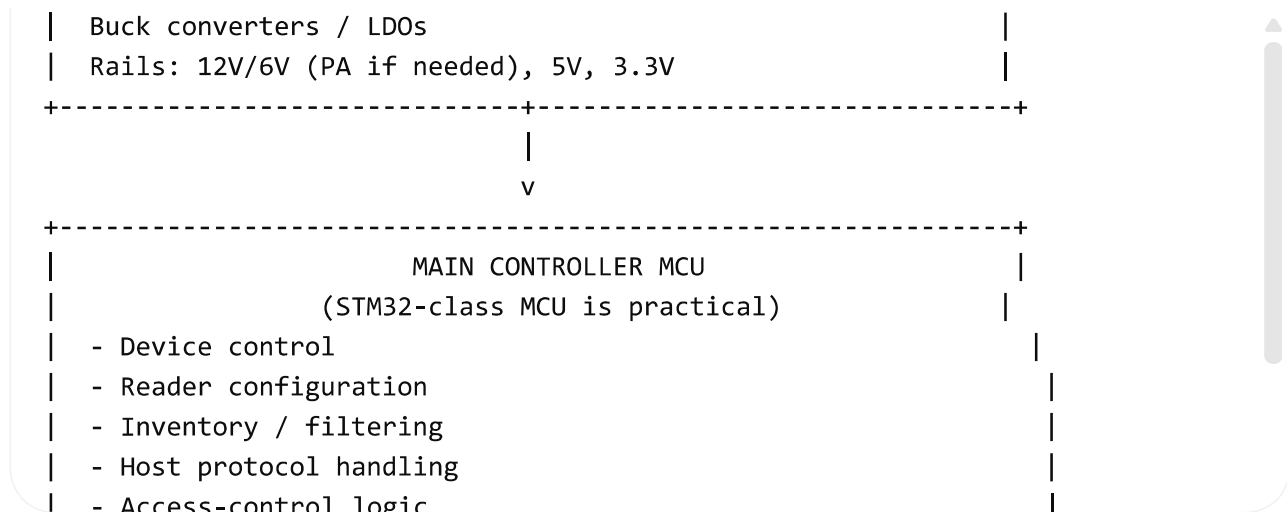
I do **not** have an official internal schematic for the HX-R01B-9. What follows is:

- **confirmed external product information** for HX-R01B-9: integrated UHF reader, **DC 9–30 V**, up to 30 dBm, TCP/IP / RS232 / RS485, optional **Wiegand 26/34**, and **ISO 18000-6C/6B** support RFIDLabel™
- plus a **reverse-engineered reference architecture** based on standard fixed UHF reader designs and ST's documented UHF reference platforms using the **ST25RU3993** reader IC and an **STM32L476** MCU on their eval/high-power boards. STMicroelectron... +3

So this is best treated as a **design blueprint for an HX-R01B-9-class product**, not a guaranteed internal teardown of that exact unit.

1) Block diagram for an HX-R01B-9-class reader





Why this block diagram is realistic

HX-R01B-9 is publicly described as an **integrated UHF reader with up to 30 dBm output, DC 9–30 V input, and TCP/IP / RS232 / RS485 / optional Wiegand**, which strongly fits the fixed-reader architecture above. RFIDLabel™

For the reader engine, ST's **ST25RU3993** is a close reference-class IC because it supports the **840–960 MHz UHF band, ISO 18000-62/-63** (the modern lineage around 6B/6C families), includes **frequency hopping, ASK / PR-ASK**, framing/CRC support, and is used by ST in a documented high-power fixed-reader system. STMicroelectron... +2

ST's high-power reader reference also explicitly uses:

- **STM32L476RGT6** as the controller
- **external PA**
- **RF switching**
- **antenna path**
- **USB/UART host control in the reference setup** STMicroelectron... +1

That makes it a good architecture model for an HX-R01B-9-class product.

2) Functional element list to design it

A. Power input section

Required elements

- DC input connector or terminal block
- reverse-polarity protection

- TVS diode for surge protection
- input fuse or resettable fuse
- buck converter from **9–30 V** to intermediate rails
- 5 V regulator
- 3.3 V regulator
- optional 6 V / PA rail depending on RF power stage

Why

HX-R01B-9 is listed with **DC 9–30 V** input, so the board needs a wide-input industrial power stage. RFIDLabel™

ST's high-power reference separately powers the **external PA**, showing that a dedicated PA rail is normal in this class of reader. STMicroelectron... +1

B. Main MCU / digital control section

Required elements

- MCU, preferably **STM32-class**
- main crystal or oscillator
- reset circuit
- SWD/JTAG programming header
- boot mode resistors
- local decoupling capacitors
- external flash or EEPROM for settings, optional

Suggested MCU class

- **STM32L4** for a proven reference path
- **STM32G4** for a more industrial control flavor
- **STM32H7** if you need richer networking/UI

Why

ST's documented UHF eval/high-power reader systems use **STM32L476** to control the reader subsystem. STMicroelectron... +1

C. UHF reader IC / analog front-end

Required elements

- UHF RFID reader IC such as **ST25RU3993** or similar class
- RF reference clock
- SPI interface wiring to MCU
- IRQ and enable lines
- RF matching around the IC
- decoupling and analog filtering

Why

The **ST25RU3993** is specifically a **UHF RFID reader device** for **840–960 MHz**, with protocol support, framing, CRC, filtering, and reader-side RF/baseband features that a raw MCU does not provide by itself. STMicroelectron... +1

D. RF transmit / receive front-end

Required elements

- impedance matching network
- balun or differential-to-single-ended conversion as needed
- RF filters
- RF switch
- directional coupler or RF detector
- receive protection / leakage management
- optional carrier-cancellation network

Optional but recommended for long range

- **external RF power amplifier**
- output power monitoring
- thermal sensing near PA
- shield can over RF section

Why

ST's high-power reader design explicitly includes an **external PA**, RF switching, and the typical fixed-reader RF signal chain around the reader IC. STMicroelectron... +1

The **ST25RU3993** itself supports modulation and receive-side features, but practical long-range designs still need board-level RF support around it. STMicroelectron... +1

E. Antenna section

Required elements

- integrated panel antenna or patch antenna
- 50 Ω feed network
- tuning/matching components
- optional antenna connector for testing
- mechanical radome/enclosure spacing

Likely antenna target

Since the family includes **HX-R01B-4 / HX-R01B-9 / HX-R01B-12**, the suffix likely indicates antenna-gain class, so HX-R01B-9 is likely a **9 dBi integrated antenna variant**, though I have not verified that exact statement from an official internal datasheet. That part remains an informed inference from the family naming. RFIDLabel™

F. Host communication interfaces

Required elements depending on your target product

- RJ45 + Ethernet PHY + magnetics for TCP/IP
- RS232 transceiver
- RS485 transceiver
- Wiegand output driver / isolation / connector
- ESD protection on each external port

Why

HX-R01B-9 is publicly listed with **TCP/IP or RS232**, plus **RS485**, and optional **Wiegand 26/34**. RFIDLabel™

G. User / service interface

Recommended elements

- status LEDs
- buzzer
- service button / reset button
- DIP switches or rotary selector for node ID / mode, optional

- debug UART header
- USB-UART bridge for development, optional

Why

These are not publicly listed as HX-R01B-9 features, but they are standard support elements in fixed-reader products and eval systems. ST's reader eval stack is controlled through a host GUI over USB/UART in its documented setup. STMicroelectron... +1

H. Mechanical / enclosure elements

Required elements

- weather-resistant enclosure if outdoor use
- RF window / plastic front
- heat path for PA if using 30 dBm-class output
- cable glands / sealed connectors
- PCB mounting frame
- grounding plan for RF and surge protection

Why

HX-R01B-9 is described as an integrated long-range fixed reader in an engineering-plastic housing with sizable enclosure dimensions, which implies enclosure and thermal design are part of the product, not an afterthought. RFIDLabel™

3) Compact BOM-style element list

Core ICs

- 1 × STM32 MCU
- 1 × UHF RFID reader IC
- 1 × Ethernet PHY
- 1 × RS232 transceiver
- 1 × RS485 transceiver
- 1 × optional Wiegand driver / interface logic
- 1 × optional external RF PA
- 1 × buck regulator controller / power IC
- 1–2 × LDOs

RF parts

- patch/panel antenna
- RF switches
- balun
- matching inductors/capacitors
- SAW/filter parts as needed
- directional coupler / detector
- coax test connector
- shield can

Protection / interface parts

- TVS diodes
- ESD arrays
- fuse / reverse-protection MOSFET
- magnetics for Ethernet
- terminal block or DC jack

Support parts

- crystals / oscillators
 - SPI pull-ups / line conditioning
 - LEDs / buzzer / buttons
 - flash / EEPROM optional
 - debug connector
-

4) Minimum design versions

Version A — simplest prototype

Build:

- STM32 MCU
- ST25RU3993-class reader IC
- one antenna path
- UART debug
- no Ethernet yet
- no Wiegand yet
- internal or modest RF output first

This is enough to prove:

- tag reading
- 6C operation
- inventory loop
- host control

Version B — HX-R01B-9-class fixed reader

Add:

- wide-input 9–30 V power stage
- Ethernet + RS232 + RS485
- optional Wiegand
- stronger RF front-end / PA
- integrated directional antenna
- outdoor mechanical design
- thermal and EMC work

That is much closer to the published HX-R01B-9 profile. RFIDLabel™ +1

5) Honest engineering caveat

The hardest part is not the STM32 firmware. It is usually:

- RF front-end tuning
- antenna design
- leakage/coupling control
- PA linearity and thermal behavior
- EMC/regulatory compliance
- read stability in real installations

That is exactly why commercial fixed readers integrate so much RF-specific hardware around the core reader chip. ST's documented reader systems make this clear by centering the design on a dedicated reader IC plus RF support circuitry, not on MCU-only implementation. STMicroelectron... +2

6) Recommended starting point

If your goal is to design something like HX-R01B-9, start with this stack:

- **STM32L476-class MCU**

- ST25RU3993-class UHF reader IC
- single antenna path
- 6C first
- RS232 first
- then add Ethernet / RS485 / Wiegand
- add external PA only after the first RF prototype is stable

That follows a documented and practical reference path. STMicroelectron... +2

I can turn this into a **clean engineering block diagram image** or a **table-form BOM with part categories and design notes** next.



I am going to design UHF long range RFID reader at
800~900 MHz range

I'm pulling current primary-source references for the RF, protocol, and compliance side so
I can give you a design path that is technically realistic, not just generic advice.

Thought for 14s

Yes — that is feasible, but the first critical decision is **which country/region you are designing for**.

A "UHF long-range RFID reader at **800–900 MHz**" is **too broad** for a real product, because passive UHF RFID is operated in **different locally approved subsets** of the broader UHF range depending on region. RAIN RFID guidance says readers must be configured for the **approved frequency band for the country/region of operation**, within a locally regulated subset of the broader **860–930 MHz** range, and that using the correct regional settings is a legal requirement. The RAIN Alliance +2

Best target definition

For a real design, define your product as one of these:

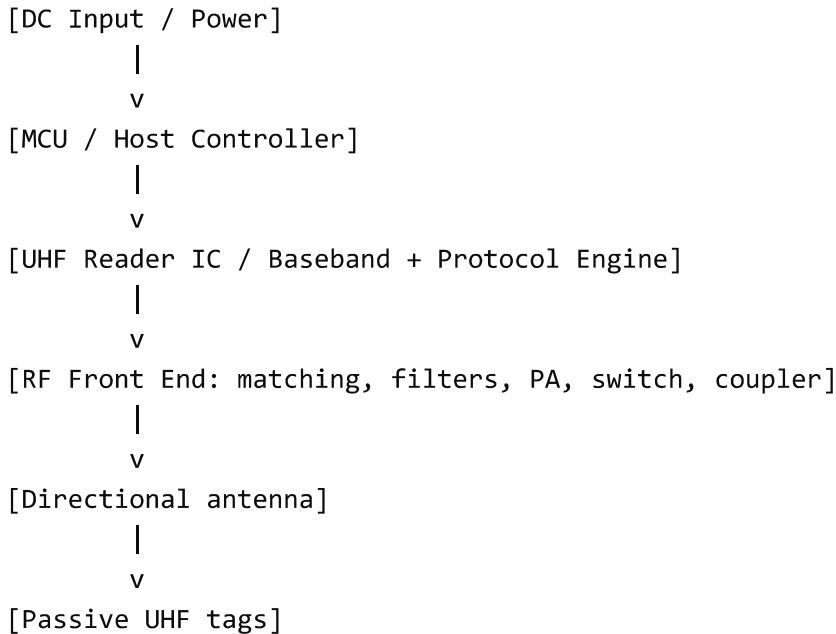
- **US-focused:** around **902–928 MHz**
- **EU-focused:** around **865–868 MHz** or current permitted RAIN sub-bands by country/region

- **multi-region SKU:** wideband front end, but region-locked firmware and compliance settings

RAIN's current radio-regulation guidance shows that UHF RFID deployments are region-specific within the global UHF space, and recent RAIN material notes continued evolution in Europe's upper-band usage. [The RAIN Alliance +2](#)

Recommended design direction

For a **long-range passive UHF reader**, the most practical architecture is:



This is the same overall pattern used by documented reader chip platforms like ST's **ST25RU3993**, which supports the **840–960 MHz** UHF band and provides protocol support, dense-reader features, framing, CRC, decode, hopping support, and an on-chip VCO and PA. ST positions it as the core of a complete reader system driven by an external microcontroller. [STMicroelectron... +1](#)

Strong recommendation: target 6C first

If this is a new design, prioritize **ISO/IEC 18000-6C / EPC Gen2 / RAIN RFID** first.

GS1 identifies EPC UHF Gen2 as the protocol defining the physical and logical requirements for UHF RFID systems, and RAIN RFID is built on the modern Gen2 standards lineage. That gives you the largest tag ecosystem, best interoperability, and the most realistic path for a new long-range reader. [GS1 +2](#)

Support for 6B is possible in some reader ICs, but it is mainly valuable for legacy compatibility. ST's UHF reader device supports **ISO18000-62 & -63, ISO29143**, and GS1

EPC UHF Gen2, and other ST material also describes **ISO18000-6c/b** support in the family context. [STMicroelectron...](#) +1

Suggested product strategy

Version 1: engineering prototype

Build a reader that can:

- read **Gen2 / 6C** passive tags reliably
- operate in **one region only**
- provide **UART or Ethernet**
- use a **single antenna port**
- output moderate power first, then scale up

This gets you to a usable prototype faster and avoids regional compliance complexity exploding too early.

Version 2: long-range fixed reader

After the prototype is stable, add:

- higher output stage
- directional antenna
- better carrier-leakage control
- region-specific channel plan / hopping behavior
- industrial interfaces like **RS-485** or **Wiegand**
- enclosure, EMC, thermal, and regulatory work

Practical chip-level architecture

A realistic build path is:

MCU

Use an **STM32** as the controller for:

- reader setup
- inventory loop
- filtering
- host comms
- diagnostics
- access-control logic

Reader IC

Use a dedicated UHF RFID reader IC such as **ST25RU3993**.

It supports:

- 840–960 MHz
- frequency hopping
- ASK / PR-ASK
- dense reader mode
- low-level transmission coding
- low-level decode
- data framing
- CRC checking
- tag read rates up to **700 tags/s** in ST's datasheet conditions STMicroelectron... +1

RF front end

For long range, you will likely need:

- matching network
- filters
- external PA
- RF switch
- coupler / detector
- carrier-cancellation or leakage control
- directional antenna

ST's product presentation for the ST25RU3993 shows long-range evaluation results reaching around **15 m** with the right tool and EIRP setup, which is a good reference point for what "long range" means in a practical passive-UHF system. STMicroelectron...

Main engineering challenges

The hard part is **not** the MCU firmware.

The hard parts are:

RF and antenna

Long-range passive RFID performance depends heavily on:

- antenna gain and pattern
- tag orientation
- polarization match
- front-end sensitivity

- self-jamming / carrier leakage
- environment and reflections

RAIN's system-design guidance and lessons-learned material both emphasize configuration, regional settings, and system-level performance effects in real deployments.

The RAIN Alliance +2

Compliance

You cannot just choose "800–900 MHz" generically and transmit there. Frequency plan, hopping, dwell behavior, power, and local approvals all depend on the region. RAIN's regulation guide is the right starting point for defining country-specific operating parameters. The RAIN Alliance +1

Reader-class performance

To get stable long-range reads, you need good:

- TX power
- RX sensitivity
- RF isolation
- antenna engineering
- firmware tuning

ST highlights **high Rx sensitivity**, **dense reader mode**, and **840–960 MHz** operation as key reasons these dedicated reader ICs exist. STMicroelectron... +1

Recommended first design spec

I would define your first project like this:

Prototype goal

Fixed passive UHF Gen2 reader for one region

Suggested baseline

- Protocol: **ISO/IEC 18000-6C / EPC Gen2**
- Region: pick **US 902–928 MHz** or **EU 865–868 MHz**
- Controller: **STM32**
- Reader IC: **ST25RU3993-class**
- Host interface: **UART first**, Ethernet later
- Antenna: **single directional antenna**
- Power: start modest, scale after RF validation

What not to do first

Do **not** start with:

- multi-region certification
- 6B + 6C + custom packet protocol all at once
- multi-antenna portals
- maximum-power PA before proving receiver stability

My honest recommendation

If your end goal is something like **HX-R01B-9**, the right sequence is:

1. choose the **target region**
2. build a **6C-only prototype**
3. validate RF front end and antenna
4. add long-range PA and industrial interfaces
5. only then consider legacy 6B or multi-region expansion

That path aligns with the current RAIN/GS1 standards ecosystem and with how modern UHF reader chip platforms are actually structured. GS1 +2

I can turn this into a **step-by-step hardware development plan** next.



give me the antenna design RF block design for it

I'm grounding this in reader-chip reference designs and antenna engineering guidance so the RF chain is something you could actually build, not just a conceptual sketch.

Thought for 32s

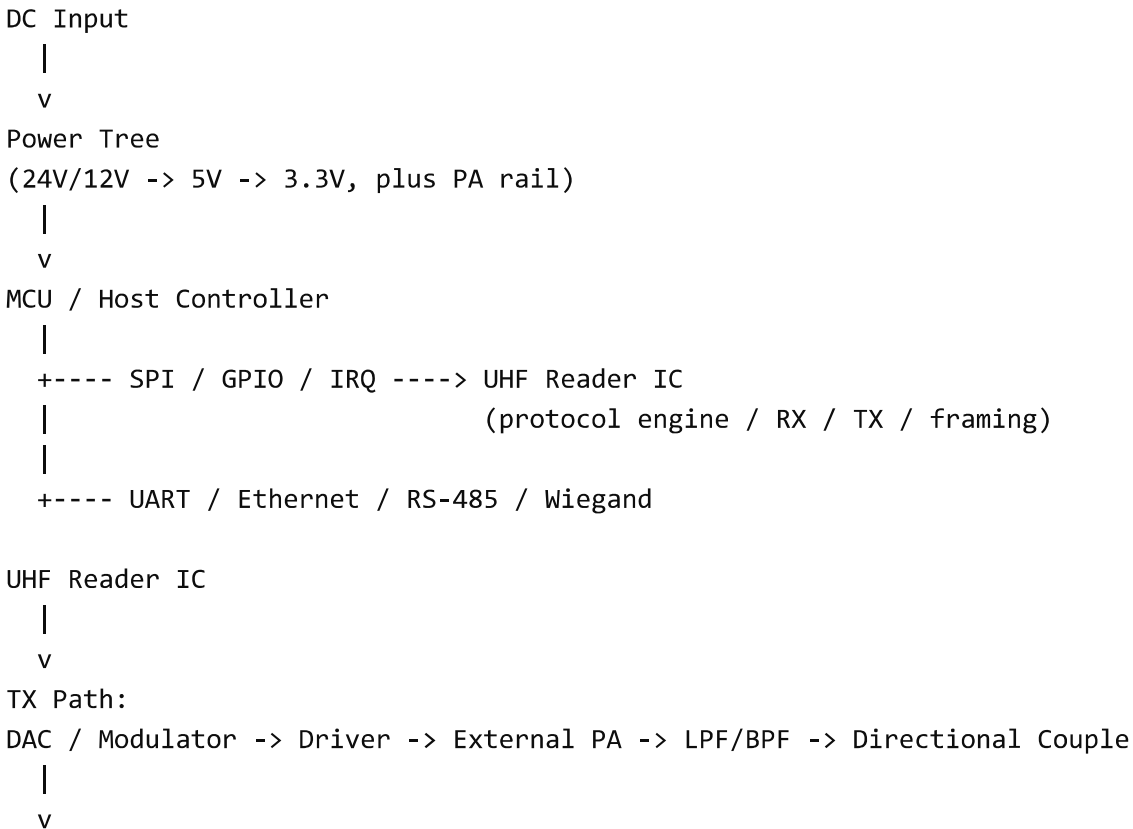
Yes. For a **long-range passive UHF RFID reader**, I would treat this as **two linked designs**:

1. the **RF reader front end**
2. the **reader antenna**

For a real product, the antenna cannot be finalized until you lock **region/band**, **EIRP target**, **polarization**, **read-zone shape**, and **mechanical size**. A reader built around a chip like **ST25RU3993** can cover **840–960 MHz**, but regional operation must still be limited to the approved band and channel plan for the target country. STMicroelectron... +1

1) Recommended RF block design

System block



That block partition is consistent with ST's high-power reference reader, which combines the **ST25RU3993**, an **external power amplifier**, antenna switching, test points, and a host MCU into one reader platform. ST also states the reader IC itself provides low-level RFID functions such as framing, CRC, hopping support, ASK/PR-ASK, and dense-reader support, which is why the external RF chain is built around it rather than replaced by the MCU.

STMicroelectron... +1

2) What each RF block should do

Reader IC / protocol engine

Use a dedicated UHF reader IC as the PHY/baseband core. The **ST25RU3993** supports the **840–960 MHz** band, **ISO 18000-6C**, and **ISO 18000-6A/B** through **direct mode**, and includes an on-chip VCO, PA support, low-level coding/decoding, data framing, CRC checking, and frequency hopping support.

STMicroelectron...

External PA

For a true long-range fixed reader, plan on an **external PA stage**. ST's high-power platform explicitly adds an external PA and reaches about **30.5 dBm max TX power** on the high-power board, versus lower output on the internal-PA path. RFID JOURNAL +1

Filters and matching

Place output filtering and matching between the PA and the antenna path. Even if your exact topology differs, you will still need:

- impedance transformation to 50 Ω
- harmonic suppression
- stable load presented to the PA
- receive-side protection from TX leakage

Those requirements follow from standard UHF reader architecture and from ST's high-power reader design including matching, switching, and measurement circuitry around the RF section. STMicroelectron... +1

Directional coupler / RF detector

Add a coupler or detector so the MCU can observe forward/reflected power and assist tuning. ST's high-power reader reference includes test and measurement features and explicitly discusses reflected-power measurement and antenna-path control. STMicroelectron...

Leakage / carrier cancellation

This is one of the most important blocks for long-range readers. ST says the reader IC is designed for high sensitivity and immunity against antenna reflection and self-jamming, and the high-power board includes architecture intended for those RF realities.

STMicroelectron... +1

3) Recommended antenna architecture

For a **long-range reader**, the usual starting point is a **directional patch/panel antenna**, not a dipole. Patch-style antennas are common in UHF RFID readers because they are low-profile and easy to integrate into a fixed enclosure, though they trade bandwidth and efficiency against size and structure choices. IJECCE +1

Best first choice

Use a **single directional patch/panel antenna** designed for your target band:

- **902–928 MHz** for the US
- **865–868 MHz** for much of Europe
- other region-specific bands only if that is your actual target market

RAIN's regulatory and system-design guidance makes clear that the operational band and channel plan are region-specific. The RAIN Alliance +1

Polarization choice

Choose polarization based on tag orientation:

Circular polarization

Best when tag orientation is unknown or varies. RAIN notes that antenna polarization affects read performance, and circularly polarized reader antennas are commonly used when tag orientation is arbitrary. However, a circularly polarized reader talking to a linearly polarized tag incurs about **3 dB best-case polarization loss**. The RAIN Alliance +2

Linear polarization

Best when tag orientation is controlled. RAIN specifically recommends using polarization intentionally when object orientation through the read zone is predictable. With linear-to-linear alignment, you avoid the fixed 3 dB CP-to-LP penalty. The RAIN Alliance +1

Practical rule

- **gate / vehicle / portal with unknown tag angle** → start with **circular polarization**
- **conveyor / fixed fixture with controlled tag angle** → use **linear polarization**

That is the most practical split from the polarization guidance and reader-antenna literature. The RAIN Alliance +2

4) Reader antenna design targets

For a first fixed-reader antenna, target these properties:

- **50 Ω input**
- **good match across your actual regional band**
- **directional pattern**
- **stable performance when mounted in the enclosure**
- **enough bandwidth to survive production tolerance and nearby-plastic effects**
- **controlled polarization**
- **low axial ratio** if circular polarization is chosen

Those are standard reader-antenna requirements, and the UHF RFID literature shows that polarization purity and bandwidth are major practical constraints in fixed-reader antenna design. NASA Technical ... +2

Gain

For long range, reader antennas are commonly in the **directional panel** class rather than omnidirectional. RAIN's regulation guide also ties allowable conducted power and antenna gain to regional compliance. In other words, do not choose gain in isolation; choose it together with your legal EIRP target. The RAIN Alliance +1

Beamwidth

Design the beamwidth around the read zone:

- narrow beam for a **gate lane / parking lane**
- wider beam for a **portal / dock door**
- very controlled beam for **single-lane access**

Beamwidth is not just a coverage question; it also affects unwanted reads and multipath behavior. RAIN's field guidance emphasizes that reflections and geometry strongly affect real deployments. The RAIN Alliance +1

5) Antenna topologies I would consider

Option A: Single-feed circularly polarized patch

This is the best first candidate for a long-range fixed reader with unknown tag orientation. CP patch antennas are a common reader-side solution for RFID because they are compact and tolerate arbitrary tag orientation better than a single linear axis. NASA Technical ... +1

Option B: Dual-linear diversity patch

Instead of always paying the CP penalty, you can use two orthogonal linear polarizations and switch or combine them. NASA's RFID reader-antenna work discusses this tradeoff directly: CP is convenient for unknown orientation, but there is power loss relative to ideal LP alignment, while polarization-diversity approaches can recover performance.

NASA Technical ...

Option C: Linear patch

Use this only when the tag orientation is predictable and controlled. It gives you better link efficiency in that controlled case. RAIN's field guidance supports this recommendation.

The RAIN Alliance +1

6) Practical RF PCB design guidance

For the RF board around the reader IC and PA:

- keep the **reader IC, PA, switch, coupler, and matching network** physically tight
- maintain controlled **50 Ω RF traces**
- separate **digital noise** from the RF section
- give the PA and RF switch a strong **ground return**
- reserve space for **tuning components**
- include **RF test ports** or at least coupler pads for lab validation
- isolate the RF section under a **shield can** on later revisions

These are standard requirements for UHF reader hardware, and ST's high-power board is explicitly built with extensive measurement access and a dedicated RF section because tuning and leakage control matter so much. STMicroelectron...

7) A realistic first-pass antenna stackup

For a practical first build, I would do this:

Reader antenna V1

- **single patch/panel antenna**
- **tuned for one region only**
- **circular polarization**
- **50 Ω feed**
- mounted behind a non-metallic front cover
- SMA/SMB test feed during development
- final integrated feed only after tuning is stable

Reader RF board V1

- STM32-class MCU
- ST25RU3993-class reader IC
- external PA
- RF switch / T-R path

- coupler + detector
- one antenna port
- UART debug first, Ethernet later

That keeps the project buildable while still being close to a commercial fixed-reader architecture. [STMicroelectron...](#) +1

8) What I cannot give exactly yet

I cannot give you a **final antenna geometry** with dimensions unless you define:

- target band, such as **902–928 MHz** or **865–868 MHz**
- substrate type and thickness
- enclosure material and antenna spacing
- polarization choice
- target gain / beamwidth
- max conducted power / EIRP target

Without those, any exact dimensions would be guesswork. The antenna's resonant size, bandwidth, and axial ratio all move with substrate and enclosure choices. That is a design fact reflected throughout the RFID antenna literature and real reader reference designs.

[URSI](#) +1

9) My recommendation

Start with this exact plan:

1. choose **US 902–928 MHz** or **EU 865–868 MHz**
2. build a **single-port RF board** around a dedicated UHF reader IC
3. use a **single-feed circularly polarized patch/panel antenna**
4. tune the antenna and RF path together on the final enclosure front plate
5. only after that consider dual-polarization or multi-antenna diversity

That is the shortest path to a real long-range passive UHF reader. [The RAIN Alliance](#) +2

Give me your **target region** and **mechanical size limit**, and I'll turn this into a **concrete antenna concept** with **feed style**, **substrate choice**, and **RF block BOM categories**.

   Sources

I choose EU 865 - 868Mhz range. give me the RF element list to design it

I'm narrowing this to the EU 865–868 MHz RF chain and pulling the parts by function so you get a buildable reader-side element list rather than a generic BOM dump.

Stopped thinking

Quick answer

I choose EU 865 - 868Mhz range. give me the RF element list and prices to design it

I'm pricing a **practical EU 865–868 MHz RF chain** around a real reader architecture, using current distributor listings so the numbers are useful for planning.

Thought for 51s

For an **EU 865–868 MHz passive UHF RFID reader**, here is a **practical RF element list** for a first prototype, with **example parts** and **current indicative prices** from distributors.

Best architecture for V1

For a new design, the cleanest RF stack is:

- UHF reader IC
- external RF front-end / PA module
- RF switch
- power detector
- band-select / harmonic filtering
- matching network
- 868 MHz antenna
- SMA test connector + coax test access

This follows the way ST positions the **ST25RU3993** as the RFID reader core for **840–960 MHz** systems, with external RF hardware used to build a complete reader. DigiKey +1

RF element list with example parts and prices

1) UHF RFID reader IC

This is the RFID PHY / protocol engine.

- **STMicroelectronics ST25RU3993-BQFT**
- Function: reader core for UHF RFID / Gen2-class systems
- Price: Mouser has a product page, but the listing shows the part as **obsolete**, and the page did not expose a stable live unit price in the snippet I could verify. Mouser Electron...

Planning note: even if you use this as the architectural reference, availability risk is real because Mouser marks it obsolete. Mouser Electron...

2) RF front-end / PA module

This is the easiest way to avoid building the whole PA/LNA/switch chain from discrete parts at first.

- **Skyworks SKY66423-11**
- Function: 860–930 MHz RF FEM
- Price:
 - \$2.71 qty 1
 - \$2.34 qty 10
 - \$2.03 qty 100 Mouser Electron... +1

This part is attractive for an EU 865–868 MHz design because its operating band covers your target. Mouser Electron... +1

3) RF switch

Useful if you split TX/RX paths or want antenna-path control.

- **pSemi PE4259-63**
- Function: RF switch
- Price from DigiKey:
 - \$0.83 qty 1
 - \$0.59 qty 10
 - \$0.4658 qty 100 DigiKey

4) RF power detector / log detector

Useful for forward/reflected-power monitoring and tuning.

- **Analog Devices ADL5513ACPZ-R7**

- Function: RF detector
- Price:
 - DigiKey: \$16.94 qty 1, \$13.445 qty 10 DigiKey
 - Mouser EU: about €14.57 qty 1, €11.57 qty 10 in the verified snippet. Mouser Electron...

5) 868 MHz antenna for first RF bring-up

For first lab validation, use a ready-made **868 MHz whip antenna** instead of designing the radiator immediately.

Good examples:

- **TE / Linx ANT-868-CW-RH**
 - 868 MHz whip antenna
 - \$15.34 qty 1, \$11.95 qty 10 on Mouser US. Mouser Electron...
- **TE / Linx ANT-868-PW-LP**
 - 868 MHz permanent-mount reduced-height whip
 - \$4.86 qty 1, \$4.42 qty 10 on Mouser US. Mouser Electron...
- **TE / Linx ANT-868-CHP-T**
 - 868 MHz chip antenna
 - €2.37 qty 1 on Mouser EU
 - better for embedded low-power nodes than for a long-range reader, but useful as a reference part. Mouser Electron...

For a **long-range reader**, the whip is mainly for development. Final products usually move to a **directional panel/patch antenna**, not a small chip antenna. That is an engineering recommendation based on reader architecture, not a quoted distributor fact.

6) SAW / band filter

You should reserve space for an **868 MHz SAW or bandpass filter** in the RF chain, especially if you use a PA. Mouser EU shows an available product category for **868–870 MHz SAW filters**, confirming this is a realistic filter class for your band, though I did not verify a single specific part/price from the snippet. Mouser Electron...

7) SMA connector / RF launch

Use an SMA test port in V1 so you can tune and measure with a VNA and spectrum analyzer. I found EU Mouser results for SMA connectors, but the returned snippet was not a sensible low-cost PCB-launch example, so I would treat connector cost as a small line item rather than overfit to that search result. A typical budget placeholder of **a few euros per connector** is reasonable, but I am not treating that as source-verified pricing here.

8) Matching network parts

You will need:

- RF inductors
- NP0/C0G capacitors
- 0 Ω jumpers
- optional attenuator pads

These are usually inexpensive discrete parts, but I did not pull live prices for each because the exact values depend on your PCB, antenna, and PA choice.

Recommended RF BOM for V1

Here is the version I would actually build first:

Block	Example part	Indicative price	
RFID reader IC	ST25RU3993-BQFT	price not cleanly verified; listing found, marked obsolete	Mouser Electron...
RF FEM / PA	SKY66423-11	\$2.71 qty 1	Mouser Electron...
RF switch	PE4259-63	\$0.83 qty 1	DigiKey
RF detector	ADL5513ACPZ-R7	\$16.94 qty 1 or about €14.57 qty 1	DigiKey +1
Development antenna	ANT-868-CW-RH	\$15.34 qty 1	Mouser Electron...
Lower-cost dev antenna	ANT-868-PW-LP	\$4.86 qty 1	Mouser Electron...
Embedded antenna option	ANT-868-CHP-T	€2.37 qty 1	Mouser Electron...

What else you should include even if I did not live-price every line

Mandatory RF support parts

- 50 Ω trace design
- PA supply decoupling
- RF choke / bias parts

- matching network footprints
- filter footprints
- coupler / detector routing
- shield-can area over the RF section

Strongly recommended lab gear hooks

- SMA TX/RX test port
 - detector output to MCU ADC
 - jumper options around filter/matching stages
 - footprints for alternative matching values
-

My honest recommendation

For EU 865–868 MHz, start with:

- one region only
- 6C / Gen2 first
- ready-made 868 MHz antenna for validation
- SKY66423-11-class RF FEM
- power detector
- RF switch
- a reader IC/reference design in the ST25RU3993 class

The main risk in the list above is the **reader IC sourcing**, because the ST25RU3993 Mouser page is marked obsolete. [Mouser Electron...](#)

Budget snapshot for the RF path only

Using the verified line items above, the RF-side prototype cost for the major active parts is roughly:

- RF FEM: **\$2.71**
- RF switch: **\$0.83**
- RF detector: **\$16.94**
- dev antenna: **\$4.86 to \$15.34**

That puts the clearly priced major RF support parts at about **\$25 to \$36, excluding** the reader IC, filter network, connector, PCB, and passive tuning parts. [Mouser Electron...](#) +4

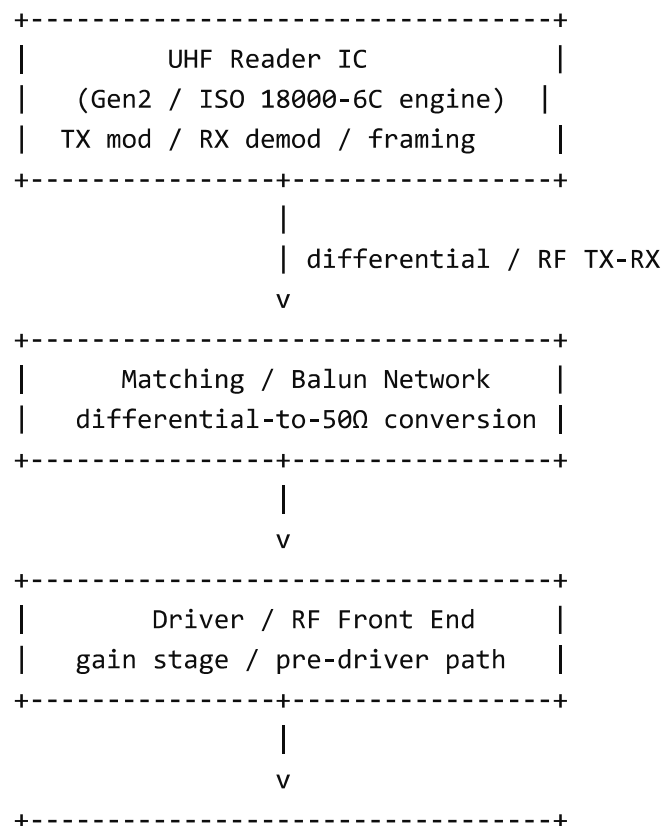
I can do the next step as a **full prototype BOM grouped by reader IC, RF, MCU, power, and interfaces** with a rough total cost.

give me the diagram for the RF block

Thought for 4s

Here is a **practical RF block diagram** for a EU 865–868 MHz long-range passive UHF RFID reader.

RF block diagram



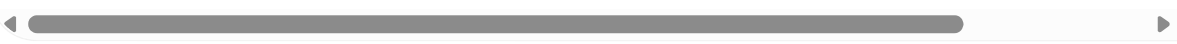
Simplified signal flow

TX path

Reader IC -> Balun/Matching -> Driver/PA -> Filter -> Coupler -> RF Switch

RX path

Tag Backscatter -> Antenna -> RF Switch -> Leakage Protection / Matching ->



What each block does

UHF Reader IC

Handles the RFID protocol side: modulation, demodulation, framing, CRC, and inventory logic support. A reader IC in the **ST25RU3993** class is designed for the **840–960 MHz** UHF RFID band and supports Gen2-class reader functions.

Matching / Balun

Converts the reader IC RF interface to a **50 Ω single-ended RF chain** and helps tune the front end.

Driver / External PA

Boosts transmit power so the reader can energize passive tags at longer range. ST's high-power reader reference uses an **external PA** for higher output fixed-reader designs.

Harmonic / Band-pass filter

Suppresses unwanted emissions and helps keep the output clean for compliance.

Directional coupler + detectors

Measures **forward** and **reflected** power so you can monitor mismatch, antenna tuning, and PA behavior.

RF switch / T-R path

Routes the antenna path and helps isolate the sensitive receive side from the stronger transmit path.

Leakage cancellation / protection

Reduces self-jamming caused by TX leakage into the RX path. This is one of the most important blocks in long-range RFID readers.

Antenna matching + reader antenna

Final tuning stage into the actual **865–868 MHz antenna**. For long-range fixed readers, a **directional patch/panel antenna** is the usual starting point.

Practical V1 version

For a first prototype, you can simplify it to this:

Reader IC

- > Matching/Balun
- > External PA
- > Band-pass filter
- > RF switch
- > Antenna matching
- > 868 MHz panel antenna

Coupler + detector taps between PA and switch

RX returns from antenna through switch back to reader IC

Recommended V1 blocks to keep

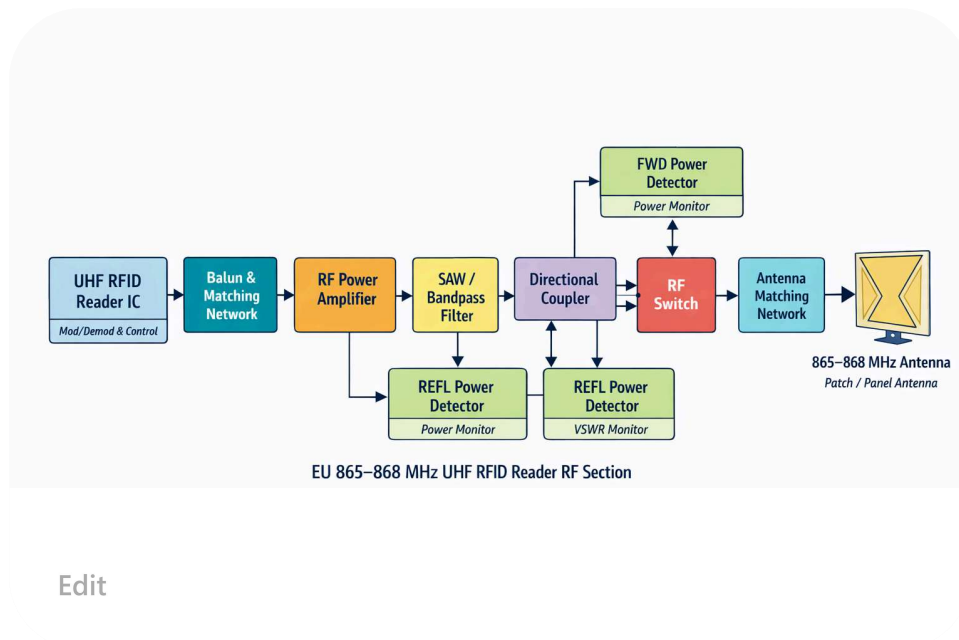
- reader IC
- balun / matching
- external PA
- filter
- coupler
- RF detector
- RF switch
- antenna matching
- one 868 MHz antenna

Optional for later revisions

- dual antenna ports
- diversity switching
- stronger leakage cancellation network
- separate LNA stage
- auto-tuning network

If you want, I can draw the **same RF block** as a cleaner engineering-style image diagram or give you a **schematic-oriented block diagram** with part labels like PA, coupler, detector, balun, and switch.

okay, please



okay, please give me more detailed explanation for each block in technically

I'm grounding the block-by-block explanation in a real reader-chip architecture so the function of each stage is technically accurate and tied to a buildable design.

Thought for 27s

Absolutely. I'll explain each RF block **technically**, in the context of a **EU 865-868 MHz passive UHF RFID reader**.

First, one important correction

In a practical reader, the signal path is usually:

Reader IC → balun/matching → PA → low-pass / band shaping filter → directional coupler → antenna switch / antenna path → final antenna matching → antenna

and the **receive path** comes back from the antenna through the switch/coupler into the **reader IC RX input**. In ST's high-power reference design, the internal or external PA output goes through an RF switch, then a low-pass filter, then a 20 dB coupler for TX-power monitoring, then a main directional coupler that separates TX and RX behavior around the antenna path. STMicroelectron... +3

1) UHF RFID Reader IC

This is the **RFID modem/baseband + RF synthesizer core** of the reader. It does not just "talk SPI." It contains the protocol-critical functions: **PLL/VCO generation**, low-level **TX coding**, **ASK / PR-ASK modulation**, **RX mixers**, **RX filtering**, **I/Q selection**, **bit decoding**, **data framing**, and **CRC checking**. On the ST25RU3993, ST explicitly lists support for **ISO 18000-6C**, **ISO 29143**, and **ISO 18000-6A/B through direct mode**, plus frequency hopping and dense-reader support in the **840–960 MHz UHF band**. STMicroelectron...

What it does electrically

The reader IC synthesizes the RF carrier and applies the modulation envelope that tags must see. On receive, it downconverts the weak tag backscatter, filters it, and reconstructs the data stream. In practice, this means the chip is handling the most timing-sensitive PHY work, while your MCU configures registers and runs the inventory/application logic. ST also notes that the chip provides **high immunity against antenna reflection and self-jamming**, which is critical in RFID readers because the receive signal is extremely small compared with the transmitted carrier. STMicroelectron...

Why it matters

This block determines:

- supported RFID air protocols
- modulation format
- receiver sensitivity ceiling
- frequency agility / hopping behavior
- how much analog RF you must add outside the chip

For EU operation, your firmware and synthesizer settings must be constrained to the proper **865–868 MHz** regional plan rather than using the whole 840–960 MHz capability of the chip. RAIN's system-design guide identifies the EU setting as based on **CEPT REC 70-03** and **EN 302 208** for **865–868 MHz**. The RAIN Alliance

2) Balun & Matching Network

This block converts the RF reader IC's output and input impedance environment into the impedance expected by the rest of the RF chain, usually **50 Ω single-ended**. In ST's reference design, the internal PA path is converted to a **50 Ω single-ended signal** after the balun, and the receive-side isolated-port signal is also transformed through a **2:1 balun** before entering the ST25RU3993 receive pins. STMicroelectron... +1

What it does electrically

A **balun** performs balanced-to-unbalanced conversion. Many RFICs present differential RF ports because that improves internal analog performance, but antennas, test gear, filters, couplers, and PA chains are generally built around single-ended 50 Ω networks. The matching part uses inductors/capacitors and sometimes transmission-line sections to:

- transform impedance
- optimize power transfer
- control return loss
- shape bandwidth
- improve stability

Why it matters

A poor balun/match will:

- reduce TX power delivered to the PA or antenna
- worsen RX sensitivity
- increase reflections
- make self-jamming harder to cancel

In a narrow EU band design, you can optimize the match specifically for **865–868 MHz**, which is easier than supporting a broad multi-region band. The RAIN Alliance +1

3) RF Power Amplifier

The PA raises the transmit signal to a level high enough to energize passive tags at useful range. In ST's high-power reference platform, the board offers two PA paths: **external PA up to 30 dBm max TX power** and **internal PA up to 17.5 dBm max TX power**, with software selecting which path is active. The internal-PA path after the balun is also described as typically around **20 dBm** before the later RF chain losses. STMicroelectron... +1

What it does electrically

The PA takes a relatively small RF drive signal and increases its power while preserving the modulation envelope enough for the tag to decode commands correctly. In RFID, linearity still matters, but not in the same way as in wideband communications; what matters most is:

- clean modulation depth
- controlled spectral emissions
- stable output under mismatch
- thermal behavior
- not desensitizing the receive path excessively

Why it matters

This block strongly affects:

- achievable read range
- compliance margin
- heat generation
- harmonic content
- required filtering after the PA

For EU 865–868 MHz, PA power cannot be chosen in isolation; it must be considered with antenna gain so the final system remains within the applicable regional constraints. RAIN's design guideline explicitly discusses EIRP and how antenna gain factors into the allowed radiated power budget. The RAIN Alliance +1

4) SAW / Bandpass / Low-pass Filter

This block cleans the PA output before it reaches the antenna system. In ST's reference design, the output of the RF switch is connected to a **low-pass filter** that attenuates the **second and third harmonics** of the carrier frequency before the signal reaches the coupler network. STMicroelectron... +1

What it does electrically

After amplification, the signal always contains unwanted spectral products:

- harmonics from nonlinearity
- switching artifacts
- spurious products from the synthesizer or modulation edges

A filter suppresses these undesired components so that:

- the transmitted spectrum is cleaner
- regulatory compliance is easier
- the antenna does not radiate too much out-of-band energy
- the detector/coupler measurements are more meaningful

Why it matters

Without proper filtering:

- compliance failures are likely
- harmonics can radiate strongly
- reflected-power measurements become less trustworthy
- the receive path may suffer more in-band and near-band contamination

For a single-region EU design, a filter optimized around **865–868 MHz** is much easier to realize than a wideband global front end. The RAIN Alliance +1

5) Directional Coupler

This is one of the most important RF blocks in a long-range reader. In ST's high-power reference, a **20 dB directional coupler** first samples a small amount of TX power for a detector, and then a **main 10 dB directional coupler** acts as the directional unit that isolates the transmit path from the receive path. ST explicitly says this main coupler plays a **very important role** because its parasitics strongly influence reader sensitivity.

STMicroelectron... +1

What it does electrically

A directional coupler is a four-port network with:

- **input**
- **direct**
- **coupled**
- **isolated**

It samples power traveling in one direction more strongly than the other. In this reader architecture it serves two major purposes:

1. TX monitoring

A small fraction of forward transmit power is sampled and sent to a detector.

2. TX/RX separation

The coupler helps route the large transmit carrier and the weak receive backscatter

differently, so the RX input is less overwhelmed by the transmitted signal.

Why it matters

This block directly affects:

- sensitivity
- self-jamming
- detector accuracy
- carrier-cancellation effectiveness

Bad coupler parasitics or poor layout here can destroy the receiver's performance even if the PA power looks good. ST says the self-jamming signal is made up of **antenna reflections (S11)** plus **leakage across the main directional coupler**, which is why this block is so critical. STMicroelectron...

6) FWD Power Detector

The forward-power detector monitors how much transmit RF power is actually flowing in the forward direction. In ST's design, the sampled TX power from the 20 dB coupler is further attenuated and then fed into a **logarithmic power detector**, whose DC output is connected to an **ADC input** of the microcontroller. ST names **MAX4003EUA** as the power detector on that board. STMicroelectron...

What it does electrically

The detector converts RF amplitude or power into a DC voltage. That voltage can be measured by the MCU to estimate actual TX power.

Why it matters

This is useful for:

- production calibration
- detecting PA failures
- closed-loop power control
- identifying mismatch conditions indirectly
- logging and diagnostics

It does **not** replace a proper RF power meter, but it gives the embedded system awareness of what the RF chain is doing. STMicroelectron...

7) REFL Power Detector

A reflected-power detector estimates how much power is coming back from the antenna side, which is a proxy for mismatch and VSWR-related problems. Your diagram shows it conceptually, and that is a sensible block even though ST's public reference emphasizes the TX detector and self-jamming/cancellation path more directly than a standalone "REFL detector" label. The reason this block is valuable is rooted in the same RF physics ST discusses: antenna reflections and coupler leakage contribute to self-jamming and reduce sensitivity. STMicroelectron...

What it does electrically

It measures RF traveling in the reverse direction relative to the intended TX flow. With a suitable coupler/detector arrangement, that can be converted to a DC value for the MCU.

Why it matters

High reflected power usually means:

- poor antenna match
- detuned enclosure/antenna integration
- cable or connector issue
- environmental loading effects

For a fixed EU reader, this is valuable during development and sometimes in the field, because the enclosure and final antenna mounting can shift the match enough to hurt range and sensitivity. This point is an engineering inference from standard coupler/detector use and ST's emphasis on reflections as a sensitivity threat. STMicroelectron...

8) RF Switch

The RF switch routes RF between alternate paths. In ST's high-power reference, one RF switch selects between the **external PA** and **internal PA** output paths, and another RF switch directs transmit power to **antenna port 1** or **antenna port 2**. ST notes that if a design only needs one RF output path, this PA-selection switch would normally be removed to avoid about **0.5 dB insertion loss**. STMicroelectron... +1

What it does electrically

An RF switch is a controlled impedance path selector. It must have:

- low insertion loss

- good isolation
- good power handling
- predictable parasitics

Why it matters

This block is used when you need:

- internal vs external PA selection
- antenna diversity
- multi-port readers
- T/R routing in some architectures

For a **single-antenna, single-PA product**, you should keep switching to a minimum because every switch adds:

- loss
- parasitics
- more return-loss problems
- more potential sensitivity degradation

That is why ST explicitly says a simpler real product would usually strip unnecessary switches out. STMicroelectron...

9) Antenna Matching Network

This is the final impedance-tuning stage between the RF chain and the actual antenna. Even if the RF front end is nominally 50 Ω , the **real antenna in the real enclosure** is almost never exactly 50 Ω over the full target band. So this block is where you absorb the practical differences between theory and the final mechanical product.

What it does electrically

Typically it consists of:

- series/shunt capacitors
- series/shunt inductors
- optional transmission-line sections
- sometimes a tunable element in development

It tunes:

- center frequency
- return loss

- bandwidth
- sometimes polarization purity indirectly if feed balance is affected

Why it matters

This stage determines whether your final antenna system is still well matched after:

- mounting behind a plastic radome
- adding screws, brackets, or shield cans nearby
- placing cables in the enclosure
- changing PCB stackup or spacing

In EU 865–868 MHz, you can optimize this matching very tightly because the operational band is narrow. RAIN's guideline identifies the EU 865–868 MHz setting as the relevant regional profile for this case. The RAIN Alliance

10) 865–868 MHz Antenna

This is the radiator that launches the RF field and receives the tag backscatter. For a long-range fixed reader, this is usually a **directional panel or patch antenna**, not a tiny chip antenna. The exact antenna choice controls the read zone at least as much as the PA does.

What it does electrically

On transmit, it converts guided RF energy into an electromagnetic field. On receive, it collects the extremely weak backscattered modulation from passive tags.

Key design parameters

For this block, the important parameters are:

- gain
- beamwidth
- polarization
- return loss
- bandwidth
- front-to-back ratio
- enclosure interaction

Why it matters

This block largely defines:

- read-zone shape

- maximum range
- unwanted side reads
- tag orientation tolerance
- final EIRP together with PA output

And in the EU band, antenna gain must be considered together with conducted transmit power because compliance is based on the radiated result, not only on the PA number. RAIN's guidance explicitly discusses EIRP and the relationship between antenna gain and transmitted power. The RAIN Alliance +1

11) Carrier Cancellation / Leakage Cancellation

Your generated image does not show this block explicitly, but it is important enough that it should be in the real design.

ST's reference design uses a **carrier cancellation circuit** connected to the **coupled port** of the main directional coupler. Its job is to reflect a controlled amount of sampled carrier back into the coupler so that it cancels the **self-jamming** signal at the isolated port before the signal enters the receiver. ST explains that if the reflected cancellation signal has the same amplitude and opposite phase as the self-jammer, they cancel. The circuit uses **three digital tunable capacitors** under **STM32 control via SPI**. STMicroelectron...

What it does electrically

It synthesizes an RF cancellation vector that reduces the residual carrier leaking into the RX path.

Why it matters

This is one of the main reasons a reader can remain sensitive while transmitting a much stronger carrier nearby. Without good cancellation, the RX chain loses dynamic range and sensitivity.

12) Receive Path Back Into the Reader IC

Even though the diagram focuses visually on the transmit chain, the real receive path is just as important. ST's reader IC contains:

- input mixers
- local oscillator path

- fast AC coupling
- RX filter
- IQ selection
- bit decoder
- data framer

These are listed directly in the datasheet functional overview. STMicroelectron...

What happens here

The antenna receives the tag's backscattered response. After coupler/switch/cancellation behavior has reduced the large residual carrier, the receive signal enters the IC, where it is downconverted, filtered, digitized/decoded, framed, and checked.

Why it matters

In passive RFID, the RX path is typically the harder side of the design:

- tag responses are weak
- TX leakage is huge by comparison
- antenna reflections vary with installation
- receive sensitivity is often what limits system performance more than raw TX power

ST's reference board quotes RX sensitivity figures down to around **-80 dBm** under specified conditions, which shows how much this block matters. STMicroelectron...

Practical summary

If I reduce the whole RF section to the essential engineering job of each block:

- **Reader IC:** makes and decodes the RFID waveform
 - **Balun/match:** converts chip ports into usable RF impedance
 - **PA:** creates enough field strength for passive tags
 - **Filter:** cleans the PA output
 - **Coupler:** samples power and helps separate TX from RX
 - **Power detectors:** give the MCU visibility into RF behavior
 - **RF switch:** selects antenna or PA path when needed
 - **Antenna match:** tunes the final installed RF path
 - **Antenna:** defines the actual read zone
 - **Carrier cancellation:** preserves RX sensitivity in the presence of strong TX leakage
-

My recommendation for your EU 865–868 MHz design

For a first real prototype, keep it as:

- one **reader IC**
- one **external PA**
- one **low-pass / band filter**
- one **main directional coupler**
- one **TX power detector**
- one **single antenna path**
- one **antenna matching stage**
- one **directional patch/panel antenna**
- add **carrier cancellation** early, not later

That is much closer to a buildable fixed reader than a simplified “reader chip + PA + antenna” concept. STMicroelectron... +3

I can turn this next into a **schematic-oriented block diagram with signal names and recommended test points**.

  Sources